

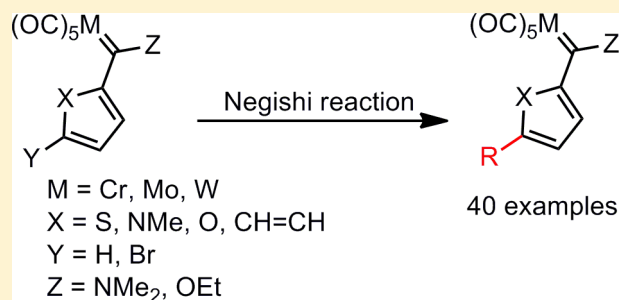
Negishi Cross-Coupling Reaction as a Simple and Efficient Route to Functionalized Amino and Alkoxy Carbene Complexes of Chromium, Molybdenum, and Tungsten

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S Supporting Information

ABSTRACT: Synthesis of functionalized phenyl, 2-thienyl, and *N*-methyl-pyrrol-2-yl Fischer carbene complexes of chromium and tungsten was achieved via a Negishi cross-coupling reaction of metalated aminocarbenes in the presence of a palladium catalyst. The reverse approach, i.e., coupling bromocarbenes with organozinc reagents, is also feasible and is particularly useful with alkoxy carbene complexes, metalation of which is not possible. Both these procedures are also suitable for the cross-coupling of molybdenum amino- and alkoxy carbene complexes. The presented methodology enables an access to the new amino and alkoxy carbene complexes bearing aldehyde, ketone, nitrile, and ester functionality.

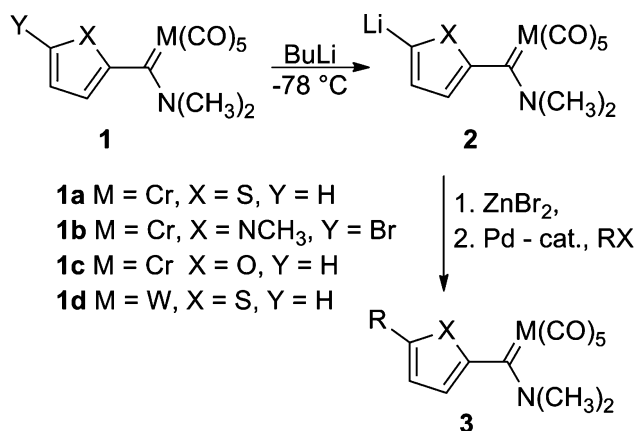


INTRODUCTION

Thermal¹ and photochemical² reactions of Fischer carbene complexes of chromium and tungsten have found a number of applications in the synthesis of complex organic molecules. Some of these reactions are moderated by other transition metals and involve transfer of a carbene ligand to the late transition metals, including copper,³ gold,⁴ nickel,^{3c,5,6} palladium^{7,8} and rhodium.^{6a,b,8c,9,10} Relatively little attention has been paid to the transition-metal-mediated modifications of already formed carbene complexes. For instance, ring-closing metathesis was used for the synthesis of complexes with attached carbocycles or heterocycles,¹¹ and Fischer biscarbene complexes were prepared by oxidative cross-coupling¹² and by click chemistry.¹³ Little is known about palladium-catalyzed C–C cross-coupling reactions of Fischer alkoxy and amino carbene complexes. To the best of our knowledge, the only reported examples are the Stille reaction of stannylated [(dimethylamino)(ethynyl)methylene]tungsten(0) carbene¹⁴ and palladium-catalyzed allylic substitution, in which an anion of chromium aminocarbene complexes serve as the nucleophile.¹⁵

Recently, we have observed that heterocyclic aminocarbene complexes **1a–d** can be smoothly lithiated using BuLi at –78 °C. The formed lithiated carbenes **2a–d** can be alkylated by primary alkyl halides and/or transformed to secondary and tertiary alcohols by the reaction with aldehydes or ketones, or acylated giving ketones.¹⁶ The feasibility of the lithiation of **1a–d** led us to explore the potential of these lithiated carbene complexes in Negishi cross-coupling reactions (Scheme 1).

Scheme 1. Negishi Cross-Coupling of Metalated Aminocarbene Complexes



RESULTS AND DISCUSSION

Due to the reported facile chromium to palladium transmetalation of the carbene moiety,^{7,8} the stability of **1a** in THF at room temperature in the presence of palladium catalysts was monitored using ¹H NMR with benzaldehyde dimethylacetal as the internal standard. Negligible decomposition was observed within 1 h, and during 24 h, the content of **1a** decreased to 87% (Pd(dba)₂) and 79% (Pd(PPh₃)₄), respectively.

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The observed stability of **1a** in the presence of Pd-catalysts encouraged us to study the ability of the lithiated carbenes **2a–c** to react with aryl and alkenyl halides under Negishi reaction conditions. Thus, the aminocarbene **1a** was treated with butyllithium at $-78\text{ }^{\circ}\text{C}$, followed by transmetalation with ZnBr_2 . The formed organozinc reagent was then reacted with 4-iodonitrobenzene under various conditions (Scheme 1, Table 1). The $\text{Pd}(\text{PPh}_3)_4$ appeared to be the best catalyst giving a full

Table 1. Negishi Cross-Coupling Reaction of Metalated Aminocarbenes **1a–d^a**

entry	1, RX	3, yield (%) ^b
1	1a , 4- $\text{NO}_2\text{C}_6\text{H}_4\text{I}$	3a , 85
2	1a , 4- $\text{NO}_2\text{C}_6\text{H}_4\text{I}$	3a , 78
3	1a , 4- $\text{NO}_2\text{C}_6\text{H}_4\text{I}$	3a , 66 ^c
4	1a , 4- $\text{NO}_2\text{C}_6\text{H}_4\text{I}$	3a , 15 ^d
5	1a , 4- $\text{NO}_2\text{C}_6\text{H}_4\text{I}$	3a , 9 ^e
6	1a , $\text{C}_6\text{H}_5\text{I}$	3b , 83
7	1a , 4- $\text{CH}_3\text{C}_6\text{H}_4\text{I}$	3c , 60
8	1a , 4- $(\text{CH}_3)_2\text{NC}_6\text{H}_4\text{I}$	3d , 64
9	1a , 2- $\text{CH}_3\text{C}_6\text{H}_4\text{I}$	3e , 74 ^f
10	1a , 4- $(\text{HCO})\text{C}_6\text{H}_4\text{I}$	3f , 74
11	1a , 4- $(\text{CH}_3\text{CO})\text{C}_6\text{H}_4\text{I}$	3g , 72
12	1a , 4- $\text{MeO}_2\text{CC}_6\text{H}_4\text{I}$	3h , 43
13	1a , 2- $\text{BrC}_5\text{H}_4\text{N}$	3i , 71
14	1a , (Z)- $\text{EtO}_2\text{CCH}=\text{CHI}$	3j , 78
15	1b , 4- $(\text{CH}_3\text{CO})\text{C}_6\text{H}_4\text{I}$	3k , 72
16	1b , 2- $\text{BrC}_5\text{H}_4\text{N}$	3l , 72
17	1c , 4- $(\text{CH}_3\text{CO})\text{C}_6\text{H}_4\text{I}$	3m , 64
18	1c , 4- $(\text{HCO})\text{C}_6\text{H}_4\text{I}$	3n , 56
19	1d , 4- $(\text{CH}_3\text{CO})\text{C}_6\text{H}_4\text{I}$	3o , 83
20	1d , 2- $\text{BrC}_5\text{H}_4\text{N}$	3p , 78

^aReaction conditions: 5 mol % of the $\text{Pd}(\text{PPh}_3)_4$, THF, ambient temperature, 3 h. ^bIsolated yield. ^c5 mol % of the $\text{Pd}(\text{dba})_2/\text{TFP}$ was used. ^d5 mol % of the $\text{Pd}(\text{dba})_2$ was used. ^e5 mol % of the $\text{Pd}(\text{tBu}_3\text{P})_2$ was used. ^fReaction time was 20 h.

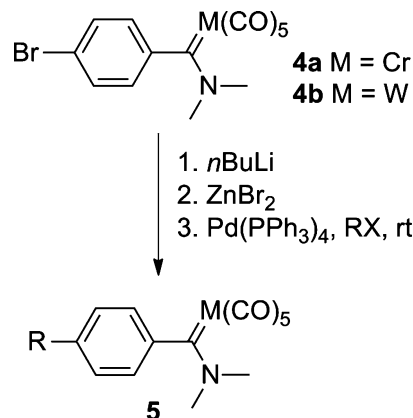
conversion and 85% isolated yield of the expected product **3a** within 3 h (Table 1, entry 1). An extension of the reaction time to 20 h slightly lowered the yield of **3a** to 78% (Table 1, entry 2). The use of $\text{Pd}(\text{dba})_2$ with tri(2-furyl)phosphine (TFP) gave only a moderate yield of **3a** (Table 1, entry 3), while the yields with $\text{Pd}(\text{dba})_2$ or $\text{Pd}(\text{t-Bu}_3\text{P})_2$ were low (Table 1, entries 4 and 5).

The best reaction conditions ($\text{Pd}(\text{PPh}_3)_4$, 3 h) were used to examine the scope of this reaction. Thus, the iodobenzene was converted to **3b** in 83% yield (Table 1, entry 6). Iodobenzenes bearing electron-donating groups yielded the expected products **3c,d** in good yields (Table 1, entries 7 and 8). In the case of 2-iodotoluene, the reaction time had to be extended to 20 h to reach the full conversion. The product **3e** was then obtained with a yield of 74% (Table 1, entry 9). Similarly, high yields were obtained with iodobenzenes containing electron-withdrawing substituents (Table 1, entries 10–12) and with 2-bromopyridine (Table 1, entry 13). The only exception was methyl-4-iodobenzoate, which gave only 43% yield of **3h** (Table 1, entry 12). The reaction with (Z)-ethyl-3-iodopropenoate furnished the corresponding carbene **3j** in 78% yield (Table 1, entry 14) with full retention of the configuration on the double bond. Pyrrole and furane-derived carbenes **1b,c** also reacted easily with 4-iodoacetophenone, 4-iodobenzaldehyde, or 2-bromopyridine under the above conditions, giving the

expected coupled products in yields ranging from 56% to 72% (Table 1, entries 15–18). This methodology, demonstrated by the reaction of **1d** with 4-iodoacetophenone and 2-bromopyridine, can be extended to the tungsten carbene complexes, furnishing the coupled products **3o,p** even in somewhat higher yields compared to their chromium analogues (Table 1, entries 19 and 20).

Starting from the easily available bromocarbene **4a**,^{8a} the developed protocol also enables the synthesis of substituted biphenyls bearing an aminocarbene moiety (Scheme 2). In this

Scheme 2. Cross-Coupling of Metalated 4-Bromophenylcarbenes **4a and **4b** with Alkyl Halides**



way, biphenylaminocarbenes bearing formyl, acetyl, and ester groups **5a–c** were obtained in 65–77% yield (Table 2, entries

Table 2. Cross-Coupling Reaction of Metalated Phenylcarbenes **4a and **4b** with Alkyl Halides Catalyzed by $\text{Pd}(\text{PPh}_3)_4$ ^a**

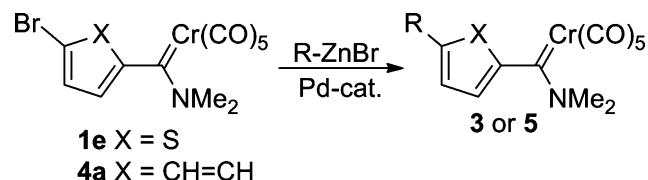
entry	4, RX	5, yield (%) ^b
1	4a , 4- $(\text{HCO})\text{C}_6\text{H}_4\text{I}$	5a , 65
2	4a , 4- $(\text{CH}_3\text{CO})\text{C}_6\text{H}_4\text{I}$	5b , 77
3	4a , 4- $\text{EtO}_2\text{CC}_6\text{H}_4\text{I}$	5c , 72
4	4a , (Z)- $\text{EtO}_2\text{CCH}=\text{CHI}$	5d , 71
5	4a , 2- $\text{BrC}_5\text{H}_4\text{N}$	5e , 85
6	4b , 4- $(\text{CH}_3\text{CO})\text{C}_6\text{H}_4\text{I}$	5f , 52
7	4b , 2- $\text{BrC}_5\text{H}_4\text{N}$	5g , 60

^aReaction conditions: 5 mol % of the catalyst, THF, ambient temperature, 3 h. ^bIsolated yield.

1–3). The reaction with (Z)-ethyl-3-iodopropenoate proceeded smoothly, furnishing **5d** in 71% yield with the retention of the double-bond configuration (Table 2, entry 4). Furthermore, 2-bromopyridine was a good substrate for this reaction, giving the coupled product **5e** in 85% yield (Table 2, entry 5). In contrast, the tungsten bromoaminocarbene **4b**¹⁶ afforded only moderate yields of the coupled products **5f,g** in the reaction with 4-iodoacetophenone and 2-bromopyridine (Table 2, entries 6 and 7).

We also focused on the reverse experimental setup of the Negishi cross-coupling reaction based on the reaction of bromocarbenes with alkyl or arylzinc bromides. The reactivity of the aminocarbene complexes **4a** and **1e** was examined first (Scheme 3). To our surprise, the $\text{Pd}(\text{PPh}_3)_4$ -catalyzed reaction of aminocarbene **1e** with butylzinc bromide, prepared by reaction of butyllithium with zinc bromide, afforded the

Scheme 3. Cross-Coupling of Bromoaminocarbenes **1e and **4a** with Organozinc Reagents**



expected product in only trace amounts. According to ^1H NMR, 30% of the unreacted **1e** remained in the reaction mixture after 3 h along with products of its decomposition. The decomposition also occurred when $\text{Pd}[\text{P}(t\text{-Bu})_3]_2$ or PEPPSI-*i*Pr were used as catalysts. However, the reaction catalyzed with $\text{PdCl}_2\text{XantPhos}$ proceeded smoothly, affording the desired product in 62% isolated yield (Table 3, entry 1). The aromatic

Table 3. Cross-Coupling Reaction of Aminocarbenes **1e and **4a** with Organozinc Reagents Catalyzed by $\text{PdCl}_2\text{XantPhos}^a$**

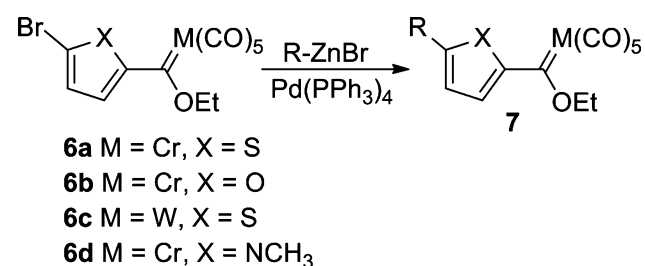
entry	bromocarbene, R	product, yield (%) ^b
1	1e , $\text{CH}_3(\text{CH}_2)_3$	3q , 62
2	1e , $4\text{-CH}_3\text{O}_2\text{CC}_6\text{H}_4$	3h , 62
3	1e , $4\text{-(CH}_3)_2\text{NC}_5\text{H}_4$	3d , 64
4	4a , $4\text{-EtO}_2\text{CC}_6\text{H}_4$	5c , 74
5	4a , 2-pyridyl	5e , 52

^aReaction conditions: 5 mol % of the catalyst, THF, ambient temperature, 3 h. ^bIsolated yield.

zinc bromides containing electron-withdrawing or -donating groups reacted similarly (Table 3, entries 2 and 3) and also did pentacarbonyl[(*N,N*-dimethylamino)(4-bromophenyl)methylene]chromium(0) (**4a**), giving the coupled products derived from 4-ethoxycarbonylphenyl and 2-pyridyl zinc bromides (Table 3, entries 4 and 5).

Next, we turned our attention to the alkoxycarbene complexes. These compounds are not compatible with organolithium reagents and, therefore, cannot be used for the preparation of the corresponding organozinc reagents. The reaction of halogenated carbene complexes thus becomes the only way to obtain the coupled products (Scheme 4).

Scheme 4. Cross-Coupling of Bromoethoxycarbenes **6a–d with Organozinc Reagents**



In this case, the alkoxycarbene complex **6a** reacted smoothly with butylzinc bromide under $\text{Pd}(\text{PPh}_3)_4$ catalysis giving the corresponding product in 67% yield (Table 4, entry 1). Using an excess of alkoxycarbene **6** in this type of cross-coupling reaction turned out to be important. If an excess of butyl zinc bromide was used, the yield of **7a** became lower than 10% (Table 4, entry 2). The reaction of phenylzinc bromide, prepared from phenyllithium, with alkoxycarbene **6a** gave the

Table 4. Negishi Reaction of Bromoalkoxycarbene Complexes **6a–c (Scheme 4)^a**

entry	bromocarbene, R	7 , yield (%) ^b
1	6a , $\text{CH}_3(\text{CH}_2)_3$	7a , 67
2	6a , $\text{CH}_3(\text{CH}_2)_3^c$	7a , <10
3	6a , C_6H_5	7b , 90
4	6a , C_6H_5^d	7b , 69
5	6a ,	7c , 100
6	6a ,	7d , 74
7	6a , $\text{C}_6\text{H}_5\text{C}\equiv\text{C}$	7e , 62
8	6a , 2- $\text{CH}_3\text{C}_6\text{H}_4$	7f , 42
9	6a , $\text{CH}_2=\text{CH}^d$	7g , 68
10	6a , $4\text{-CO}_2\text{EtC}_6\text{H}_4^d$	7h , 51
11	6a , $4\text{-(4-NCC}_6\text{H}_4)\text{C}_6\text{H}_4^d$	7i , 76
12	6b , C_6H_5^d	7j , 94
13	6b , $\text{C}_6\text{H}_5\text{C}\equiv\text{C}$	7k , 72
14	6c , C_6H_5^d	7l , 86
15	6c , $\text{C}_6\text{H}_5\text{C}\equiv\text{C}$	7m , 51

^aReaction conditions: 1.0 equiv of RZnBr prepared from RLi and ZnBr_2 , $\text{Pd}(\text{PPh}_3)_4$ (5 mol %), THF, ambient temperature, 3 h.

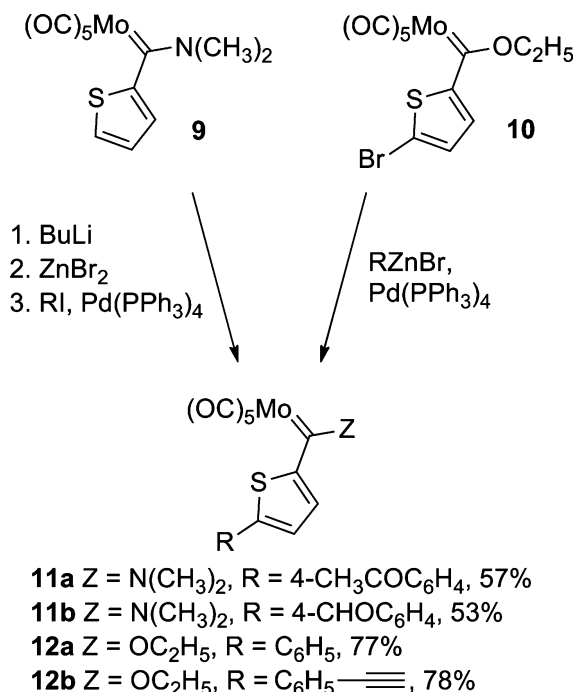
^bIsolated yield. ^c1.1 equiv of butylzinc bromide was used. ^dArylzinc bromide was prepared by reaction of arylmagnesium chloride with ZnBr_2 .

expected product **7b** in 90% isolated yield (Table 4, entry 3). In contrast, phenylzinc bromide, generated by the reaction of phenylmagnesium chloride with ZnBr_2 , furnished **7b** in 69% yield (Table 4, entry 4). In addition, other organozinc reagents derived from 2-bromo-6-methoxynaphthalene, 5-chlorothiophene and phenylacetylene were successfully coupled with alkoxycarbene **6a** (Table 4, entries 5–7). In the case of 2-iodotoluene, the reaction required 20 h to achieve full conversion and 42% yield (Table 4, entry 8). 5-Vinylalkoxycarbene **7g** was prepared starting from vinylmagnesium chloride in 68% yield (Table 4, entry 9). The recently introduced chemistry of the Turbo Grignard reagent¹⁷ enabled the preparation of functionalized Grignard reagents derived from ethyl 4-iodobenzoate and 4'-iodobiphenyl-4-carbonitrile. The subsequent transmetalation with ZnBr_2 and reaction with **6a** catalyzed by $\text{Pd}(\text{PPh}_3)_4$ led to the functionalized alkoxycarbenes **7h** and **7i** in good yields (Table 4, entries 10 and 11). The furan-derived chromium alkoxycarbene **6b** and the tungsten alkoxycarbene complex of thiophene **6c** reacted smoothly with organozinc reagents giving the corresponding coupled products (Table 4, entries 12–15). In contrast to other alkoxycarbene complexes, the pyrrole-based carbene **6d** failed to give the expected product of the cross-coupling reaction. ^1H NMR analysis along with HR MS of the crude reaction mixture confirmed that Pd -catalyzed dimerization of alkoxycarbenes occurred. Similarly, attempts to use pentacarbonyl[(ethoxy)(4-bromophenyl)methylene]chromium (**8**) as a cross-coupling

partner in the reaction with butylzinc bromide failed. Similarly, the analysis of the crude reaction mixture showed the formation of a complex mixture in which the product of self-dimerization of **8** was detected by mass spectrometry. These findings are in accordance with the reported ability of aryl alkoxy-carbenes to undergo Pd-catalyzed self-dimerization.^{7c}

As Scheme 5 shows, the cross-coupling methodology has been also successfully used for the derivatization of both alkoxy-

Scheme 5. Cross-Coupling of Molybdenum Carbene Complexes



and aminocarbene complexes of molybdenum. The obtained results are comparable with that obtained with those chromium and tungsten carbenes.

CONCLUSION

The aminocarbene complexes of chromium and tungsten bearing 2-thienyl, *N*-methyl-5-bromopyrrol-2-yl, or 4-bromophenyl groups can be, after lithiation with butyllithium at -78°C and transmetalation with zinc bromide, used in Pd-catalyzed Negishi reactions with aryl, heteroaryl, and alkenyl halides. The reverse approach, i.e., the reaction of halogenated carbene complexes with alkyl, alkenyl, aryl, and heteroaryl zinc bromides, can be used not only for the coupling of aminocarbenes but also for alkoxy-carbene complexes, lithiation of which is problematic. Thus, besides aminocarbene complexes of chromium bearing 5-bromo-2-thienyl (**1e**) and 4-bromophenyl (**4a**) groups, [ethoxy(5-bromo-2-thienyl)methylene]-chromium (**6a**), [ethoxy(5-bromo-2-furyl)methylene]-chromium (**6b**), and [ethoxy(5-bromo-2-thienyl)methylene]-tungsten (**6c**) were also successfully coupled with organozinc reagents. The molybdenum amino- and alkoxy-carbene complexes derived from thiophene can be also successfully subjected to the above-discussed cross-coupling procedures. In contrast, pentacarbonyl[ethoxy(*N*-methyl-5-bromo-2-pyrrolyl)-methylene]chromium (**6d**) and pentacarbonyl[(ethoxy)(4-bromophenyl)methylene]chromium (**8**) failed to give the coupled products and afforded only products of dimerization

of the carbene ligands. The cross-coupling reactions presented here allow the preparation of a number of new functionalized aminocarbene and alkoxy-carbene complexes of chromium and tungsten that are not easily accessible by other methods.

EXPERIMENTAL SECTION

General Information. All reactions were performed under an argon atmosphere. The solvents were dried and degassed by standard procedures; silica gel (Merck, Geduran Si 60, 63–200 μm) was used for column chromatography. Pentacarbonyl[(*N,N*-dimethylamino)(2-thienyl)carbene]chromium(0) (**1a**),¹⁸ pentacarbonyl[(*N,N*-dimethylamino)(5-bromo-1-methyl-2-pyrrolyl)carbene]chromium(0) (**1b**),¹⁶ pentacarbonyl[(*N,N*-dimethylamino)(2-furyl)carbene]-chromium(0) (**1c**),¹⁸ pentacarbonyl[(*N,N*-dimethylamino)(2-thienyl)carbene]tungsten(0) (**1d**),¹⁶ pentacarbonyl[(*N,N*-dimethylamino)(4-bromophenyl)carbene]chromium(0) (**4a**),^{8a} pentacarbonyl[(*N,N*-dimethylamino)(4-bromophenyl)carbene]-tungsten(0) (**4b**),¹⁶ pentacarbonyl[(*N,N*-dimethylamino)(5-bromo-2-thienyl)carbene]chromium(0) (**1e**), and ¹⁶ pentacarbonyl[(ethoxy)(4-bromophenyl)methylene]chromium (**8**)^{8a} were prepared by the reported procedures; other compounds were purchased.

General Procedure A for the Pd-Catalyzed Negishi Cross-Coupling Reaction of Zincated Fischer Aminocarbene Complexes. Butyllithium (1.3 equiv) was added to a solution of carbene complex **1a–d**, **4a,b** or **9** (1.0 equiv) in dry THF (10 mL/mmol) cooled to -78°C . The resultant mixture was stirred for 30 min at -78°C followed by addition of the solution of ZnBr₂ (1.0 equiv) in dry THF (0.5 M solution). After 30 min at -78°C , a solution of halide (2.0 equiv) and Pd(PPh₃)₄ (5 mol %) in dry THF (2 mL/mmol) was added. The resultant mixture was warmed to ambient temperature and stirred for 3 h. The mixture was then quenched with brine, the organic layer was separated, and the water layer was extracted with ether. Combined organic layers were dried (MgSO₄) and concentrated *in vacuo*, and the crude product was subjected to column chromatography on silica gel.

General Procedure B for the Cross-Coupling Reaction of Halogenated Amino and Alkoxy-carbenes with Organozinc Reagents. The Preparation of the Organozinc Reagents via Li/Zn Transmetalation. Commercially unavailable alkenyl-, alkynyl-, or aryllithium reagents were prepared by the reaction of BuLi (1.1 equiv) with a solution of aryl- or alkenylhalides or phenylacetylene (1.0 equiv) in dry THF (4 mL/mmol) cooled to -78°C . The resultant mixture was then stirred for 30 min at -78°C , followed by addition of the solution of freshly dried ZnBr₂ (1.0 equiv) in dry THF (2 mL/mmol). After 30 min at -78°C , the formed organozinc reagent was used in the cross-coupling reaction.

The Preparation of the Organozinc Reagent via Mg/Zn Transmetalation. Commercially unavailable alkylmagnesium halides reagents were prepared by reaction of iPrMgCl·LiCl (1.05 equiv) with a solution of arylhalides (1.0 equiv) in dry THF (4 mL/mmol) cooled to -40°C . The resultant mixture was then stirred for 30 min at -40°C , followed by addition of the solution of freshly dried ZnBr₂ (1.0 equiv) in dry THF (2 mL/mmol). After 30 min at -40°C , the formed organozinc reagent was used in the cross-coupling reaction.

The solution of alkoxy-carbene **6a–c**, **10** (1.1 equiv) and Pd(PPh₃)₄ (5 mol %) in dry THF (2 mL/mmol) was added to the freshly prepared solution of organozinc reagent (1.0 equiv) cooled to -78°C . The resultant mixture was stirred for 3 h at ambient temperature. In the case of aminocarbene complexes, a solution of aminocarbene **1e** or **4a** (1.0 equiv) and PdCl₂XantPhos (5 mol %) was added to the solution of organozinc reagent (1.2 equiv). The resultant mixture was stirred for 3 h at ambient temperature and quenched with brine, and the water layer was extracted with ether (3 × 30 mL). Combined organic layers were dried (MgSO₄) and concentrated *in vacuo*, and column chromatography (silica gel) afforded the product.

Pentacarbonyl[(*N,N*-dimethylamino)[5-(4-nitrophenyl)-2-thienyl]methylene]chromium(0) (3a**).** The general procedure A starting from **1a** (0.100 g, 0.30 mmol), BuLi (0.16 mL, 0.39 mmol), ZnBr₂ (0.068 g, 0.30 mmol), 4-nitro-1-iodobenzene (0.149 g, 0.60 mmol) and Pd(PPh₃)₄ (0.017 g, 0.015 mmol) followed by column

chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.115 g of the title compound (85%) as an orange solid. Mp: >120 °C dec. ^1H NMR (CDCl_3): δ 3.32 (s, 3H, CH_3), 4.03 (s, 3H, CH_3), 6.56 (d, J = 3.6 Hz, 1H, CH), 7.37 (d, J = 3.6 Hz, 1H, CH), 7.69 (d, J = 9.0 Hz, 2H, CH), 8.24 (d, J = 9.0 Hz, 2H, CH). ^{13}C NMR ($\text{DMSO}-d_6$): δ 47.3, 51.8, 120.4, 124.8, 125.8, 127.2, 139.5, 139.7, 146.3, 154.5, 216.8, 223.8, 258.6. IR (ATR): ν 2053 (m), 1937 (m), 1906 (s), 1882 (s), 1591 (w), 1548 (w), 1513 (m), 1449 (w), 1401 (w), 1327 (m), 1263 (w), 1214 (w), 1164 (w), 1107 (w) cm^{-1} . Anal. Calcd for $\text{C}_{18}\text{H}_{12}\text{CrN}_2\text{O}_7\text{S}$: C, 47.79; H, 2.67; N, 6.19; S, 7.09. Found: C, 47.51; H, 2.67; N, 6.03; S, 6.83.

Pentacarbonyl[(*N,N*-dimethylamino)(5-phenyl-2-thienyl)methylene]chromium(0) (3b). The general procedure A starting from **1a** (0.150 g, 0.45 mmol), BuLi (0.24 mL, 0.59 mmol), ZnBr_2 (0.101 g, 0.45 mmol), iodobenzene (0.184 g, 0.9 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.026 g, 0.023 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2) afforded 0.152 g of the title compound (83%) as a yellow solid. Mp: >81 °C dec. ^1H NMR (CDCl_3): δ 3.31 (s, 3H, CH_3), 4.00 (s, 3H, CH_3), 6.51 (d, J = 3.9 Hz, 1H, CH), 7.21 (d, J = 3.9 Hz, 1H, CH), 7.31 (m, 1H, CH), 7.38 (m, 2H, CH), 7.58 (m, 2H, CH). ^{13}C NMR (CDCl_3): δ 46.7, 51.4, 119.3, 122.8, 125.5, 127.8, 129.0, 133.5, 143.2, 151.8, 216.8, 223.6, 269.4. IR (ATR): ν 2052 (w), 1974 (w), 1886 (s), 1716 (w), 1599 (w), 1525 (w), 1492 (w), 1453 (w), 1397 (w), 1255 (w), 1222 (w), 1159 (w), 1104 (w) cm^{-1} . Anal. Calcd for $\text{C}_{18}\text{H}_{13}\text{CrNO}_5\text{S}$: C, 53.07; H, 3.22; N, 3.44. Found: C, 53.18; H, 3.51; N, 3.32.

Pentacarbonyl[(*N,N*-dimethylamino)[5-(4-methylphenyl)-2-thienyl]methylene]chromium(0) (3c). The general procedure A starting from **1a** (0.150 g, 0.45 mmol), BuLi (0.24 mL, 0.59 mmol), ZnBr_2 (0.101 g, 0.45 mmol), 4-iodotoluene (0.196 g, 0.9 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.026 g, 0.023 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.114 g of the title compound (60%) as a yellow solid. Mp: >100 °C dec. ^1H NMR (CDCl_3): δ 2.38 (s, 3H, CH_3), 3.30 (s, 3H, CH_3), 4.00 (s, 3H, CH_3), 6.49 (d, J = 3.6 Hz, 1H, CH), 7.16 (d, J = 3.6 Hz, 1H, CH), 7.19 (d, J = 8.1 Hz, 2H, CH), 7.47 (d, J = 7.8 Hz, 2H, CH). ^{13}C NMR (CDCl_3): δ 21.2, 46.7, 51.4, 119.3, 122.2, 125.5, 129.6, 130.7, 137.7, 143.5, 151.3, 216.8, 223.6, 269.3. IR (ATR): ν 2918 (w), 2051 (m), 1879 (s), 1521 (w), 1444 (w), 1398 (w), 1256 (w), 1211 (w), 1171 (w), 1102 (w), 1049 (w) cm^{-1} . Anal. Calcd for $\text{C}_{19}\text{H}_{15}\text{CrNO}_5\text{S}$: C, 54.16; H, 3.59; N, 3.32. Found: C, 54.11; H, 3.63; N, 3.32.

Pentacarbonyl[(*N,N*-dimethylamino)[5-(4-(*N',N'*-dimethylamino)phenyl)-2-thienyl]methylene]chromium(0) (3d). The general procedure A starting from **1a** (0.150 g, 0.45 mmol), BuLi (0.37 mL, 0.59 mmol), ZnBr_2 (0.101 g, 0.45 mmol), *N,N*-dimethyl-4-iodoaniline (0.168 g, 0.68 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.026 g, 0.023 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.130 g of the title compound (64%) as a yellow solid. Mp: >85 °C dec. Alternatively, the title compound was prepared following the general procedure B starting from **1d** (0.340 g, 0.83 mmol), BuLi (0.43 mL, 1.05 mmol), ZnBr_2 (0.225 g, 1.0 mmol), *N,N*-dimethyl-4-bromoaniline (0.200 g, 1.0 mmol) and $\text{PdCl}_2\text{XantPhos}$ (0.031 g, 0.042 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) giving 0.240 g of **3d** (64%). ^1H NMR (CDCl_3): δ 2.99 (s, 6H, CH_3), 3.31 (s, 3H, CH_3), 3.98 (s, 3H, CH_3), 6.47 (d, J = 3.6 Hz, 1H, CH), 6.73 (d, J = 8.4 Hz, 2H, CH), 7.03 (d, J = 3.6 Hz, 1H, CH), 7.46 (d, J = 8.4 Hz, 2H, CH). ^{13}C NMR (CDCl_3): δ 40.3, 46.7, 51.3, 112.4, 119.5, 120.3, 121.6, 126.5, 144.6, 149.9, 150.2, 216.9, 223.7, 269.1. IR (ATR): ν 2052 (m), 1972 (w), 1883 (s), 1605 (m), 1507 (m), 1438 (m), 1396 (m), 1362 (m), 1259 (w), 1226 (w), 1191 (w), 1168 (w), 1103 (w) cm^{-1} . Anal. Calcd for $\text{C}_{20}\text{H}_{18}\text{CrN}_2\text{O}_5\text{S}$: C, 53.33; H, 4.03; N, 6.22; S, 7.12. Found: C, 53.78; H, 4.23; N, 6.04; S, 7.50.

Pentacarbonyl[(*N,N*-dimethylamino)[5-(2-methylphenyl)-2-thienyl]methylene]chromium(0) (3e). The general procedure A starting from **1a** (0.150 g, 0.45 mmol), BuLi (0.24 mL, 0.59 mmol), ZnBr_2 (0.101 g, 0.45 mmol), 2-iodotoluene (0.196 g, 0.9 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.026 g, 0.023 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2) afforded 0.140 g of the title compound (74%) as a yellow oil. ^1H NMR (CDCl_3): δ 2.45 (s, 3H,

CH_3), 3.35 (s, 3H, CH_3), 4.01 (s, 3H, CH_3), 6.53 (d, J = 3.6 Hz, 1H, CH), 6.94 (d, J = 3.6 Hz, 1H, CH), 7.26–7.29 (m, 3H, CH_3), 7.40–7.42 (m, 1H, CH). ^{13}C NMR (CDCl_3): δ 21.1, 46.7, 51.3, 118.3, 125.9, 126.1, 128.1, 130.2, 130.8, 133.0, 136.2, 142.1, 152.7, 216.8, 223.7, 269.3. IR (ATR): ν 2052 (m), 1984 (m), 1882 (s), 1616 (m), 1517 (m), 1460 (m), 1395 (m), 1221 (w), 1154 (w), 1098 (w), 1037 (w) cm^{-1} . Anal. Calcd for $\text{C}_{19}\text{H}_{15}\text{CrNO}_5\text{S}$: C, 54.16; H, 3.59; N, 3.32; S, 7.61. Found: C, 54.64; H, 3.66; N, 3.36; S, 8.05.

Pentacarbonyl[(*N,N*-dimethylamino)[5-(4-formylphenyl)-2-thienyl]methylene]chromium(0) (3f). The general procedure A starting from **1a** (0.150 g, 0.45 mmol), BuLi (0.24 mL, 0.59 mmol), ZnBr_2 (0.101 g, 0.45 mmol), 4-iodobenzaldehyde (0.248 g, 0.9 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.026 g, 0.023 mmol) followed by column chromatography (silica gel, DCM/hexane, 1/2 to 2/1) afforded 0.145 g of the title compound (74%) as an orange solid. Mp: >110 °C dec. ^1H NMR (CDCl_3): δ 3.31 (s, 3H, CH_3), 4.01 (s, 3H, CH_3), 6.54 (d, J = 3.9 Hz, 1H, CH), 7.35 (d, J = 3.9 Hz, 1H, CH), 7.70 (d, J = 8.1 Hz, 2H, CH), 7.87 (d, J = 8.1 Hz, 2H, CH), 9.98 (s, 1H, CH). ^{13}C NMR (CDCl_3): δ 46.8, 51.4, 119.5, 125.0, 125.6, 130.4, 135.2, 139.3, 141.2, 153.5, 191.2, 216.6, 223.4, 269.0. IR (ATR): ν 2053 (w), 1973 (w), 1888 (w), 1691 (m), 1600 (m), 1541 (w), 1451 (w), 1399 (w), 1307 (w), 1261 (w), 1210 (w), 1164 (w) cm^{-1} . Anal. Calcd for $\text{C}_{19}\text{H}_{13}\text{CrNO}_6\text{S}$: C, 52.42; H, 3.01; N, 3.22; S, 7.36. Found: C, 51.97; H, 3.06; N, 3.12; S, 7.36.

Pentacarbonyl[(*N,N*-dimethylamino)[5-(4-acetylphenyl)-2-thienyl]methylene]chromium(0) (3g). The general procedure A starting from **1a** (0.150 g, 0.45 mmol), BuLi (0.24 mL, 0.59 mmol), ZnBr_2 (0.101 g, 0.45 mmol), 4-iodoacetophenone (0.221 g, 0.9 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.026 g, 0.023 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.146 g of the title compound (72%) as an orange solid. Mp: >92 °C dec. ^1H NMR (CDCl_3): δ 2.60 (s, 3H, CH_3), 3.30 (s, 3H, CH_3), 4.00 (s, 3H, CH_3), 6.53 (d, J = 3.9 Hz, 1H, CH), 7.31 (d, J = 3.6 Hz, 1H, CH), 7.63 (d, J = 8.4 Hz, 2H, CH), 7.95 (d, J = 8.7 Hz, 2H, CH). ^{13}C NMR (CDCl_3): δ 26.5, 46.8, 51.4, 119.4, 124.5, 125.2, 129.1, 135.8, 137.9, 141.4, 153.1, 197.2, 216.6, 223.5, 268.7. IR (ATR): ν 2051 (m), 1975 (w), 1949 (m), 1909 (s), 1878 (s), 1673 (m), 1598 (m), 1539 (w), 1451 (w), 1398 (w), 1355 (w), 1254 (w), 1215 (w), 1177 (w), 1101 (w) cm^{-1} . Anal. Calcd for $\text{C}_{20}\text{H}_{15}\text{CrNO}_6\text{S}$: C, 53.45; H, 3.47; N, 3.12. Found: C, 53.10; H, 3.63; N, 2.89.

Pentacarbonyl[(*N,N*-dimethylamino)[5-(4-methoxycarbonylphenyl)-2-thienyl]methylene]chromium(0) (3h). The general procedure A starting from **1a** (0.150 g, 0.45 mmol), BuLi (0.24 mL, 0.59 mmol), ZnBr_2 (0.101 g, 0.45 mmol), methyl 4-iodobenzoate (0.236 g, 0.9 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.026 g, 0.023 mmol) followed by column chromatography (silica gel, DCM/hexane, 1/2 to 2/1) afforded 0.130 g of the title compound (43%) as a yellow solid. Mp: >90 °C dec. Alternatively, the title compound was prepared following the general procedure B starting from **1d** (0.340 g, 0.83 mmol), *i*PrMgCl-LiCl (0.85 mL, 1.05 mmol), ZnBr_2 (0.225 g, 1.0 mmol), methyl 4-iodobenzoate (0.262 g, 1.0 mmol) and $\text{PdCl}_2\text{XantPhos}$ (0.031 g, 0.042 mmol) giving 0.241 g of the **3h** (62%). ^1H NMR (CDCl_3): δ 3.30 (s, 3H, CH_3), 3.92 (s, 3H, CH_3), 4.00 (s, 3H, CH_3), 6.52 (d, J = 3.3 Hz, 1H, CH), 7.30 (d, J = 3.3 Hz, 1H, CH), 7.61 (d, J = 8.4 Hz, 2H, CH), 8.03 (d, J = 8.1 Hz, 2H, CH). ^{13}C NMR (CDCl_3): δ 46.8, 51.4, 52.1, 119.4, 124.4, 125.1, 129.0, 130.3, 137.8, 141.7, 153.0, 166.5, 216.7, 223.5, 269.2. IR (ATR): ν 2052 (m), 1982 (w), 1881 (w), 1712 (m), 1600 (w), 1529 (w), 1432 (w), 1401 (w), 1281 (w), 1256 (w), 1215 (w), 1178 (w), 1098 (w) cm^{-1} . Anal. Calcd for $\text{C}_{20}\text{H}_{15}\text{CrNO}_7\text{S}$: C, 51.61; H, 3.25; N, 3.01. Found: C, 51.50; H, 3.31; N, 2.77.

Pentacarbonyl[(*N,N*-dimethylamino)[5-(2-pyridyl)-2-thienyl]methylene]chromium(0) (3i). The general procedure A starting from **1a** (0.150 g, 0.45 mmol), BuLi (0.24 mL, 0.59 mmol), ZnBr_2 (0.101 g, 0.45 mmol), 2-bromopyridine (0.142 g, 0.9 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.026 g, 0.023 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2) afforded 0.135 g of the title compound (71%) as a yellow solid. Mp: >114 °C dec. ^1H NMR (CDCl_3): δ 3.28 (s, 3H, CH_3), 3.98 (s, 3H, CH_3), 6.54 (d, J = 3.6 Hz, 1H, CH), 7.11–7.15 (m, 1H, CH), 7.44 (d, J = 3.6 Hz, 1H,

CH), 7.90–7.67 (m, 2H, CH), 8.51–8.52 (m, 1H, CH). ^{13}C NMR (CDCl_3): δ 46.7, 51.3, 118.4, 119.3, 122.0, 124.2, 136.6, 143.4, 149.5, 152.0, 154.2, 216.7, 223.6, 269.5. IR (ATR): ν 2054 (m), 1980 (w), 1890 (s), 1581 (m), 1520 (m), 1469 (m), 1431 (m), 1394 (m), 1294 (w), 1218 (w), 1153 (w), 1093 (w) cm^{-1} . Anal. Calcd for $\text{C}_{17}\text{H}_{12}\text{CrN}_2\text{O}_5\text{S}$: C, 50.00; H, 2.96; N, 6.86; S, 7.85. Found: C, 49.58; H, 2.96; N, 6.70; S, 7.89.

Pentacarbonyl[(Z)-{(N,N-dimethylamino)[5-(2-ethoxycarbonylvinyl)-2-thienyl]methylene}chromium(0) (3j)]. The general procedure A starting from **1a** (0.100 g, 0.30 mmol), BuLi (0.16 mL, 0.39 mmol), ZnBr_2 (0.068 g, 0.30 mmol), (Z)-ethyl 3-iodopropenoate (0.136 g, 0.6 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.017 g, 0.015 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2) afforded 0.100 g of the title compound (78%) as an orange oil. ^1H NMR (CDCl_3): δ 1.31 (t, $J = 6.9$ Hz, 3H, CH_3), 3.26 (s, 3H, CH_3), 3.98 (s, 3H, CH_3), 4.23 (q, $J = 6.9$ Hz, 2H, CH_2), 5.73 (d, $J = 12.3$ Hz, 1H, CH), 6.55 (d, $J = 3.8$ Hz, 1H, CH), 7.04 (d, $J = 12.3$ Hz, 1H, CH), 7.27 (d, $J = 3.8$ Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 14.3, 46.9, 51.3, 60.3, 113.1, 118.0, 136.1, 136.5, 136.7, 158.2, 166.6, 216.6, 223.5, 269.3. IR (ATR): ν 2054 (m), 1978 (m), 1903 (s), 1877 (s), 1698 (m), 1601 (m), 1541 (w), 1509 (w), 1452 (w), 1411 (w), 1386 (w), 1178 (m), 1097 (w), 1043 (w), 1032 (w) cm^{-1} . Anal. Calcd for $\text{C}_{17}\text{H}_{15}\text{CrNO}_7\text{S}$: C, 47.55; H, 3.52; N, 3.26; S, 7.47. Found: C, 47.84; H, 3.80; N, 3.00; S, 6.91.

Pentacarbonyl[(N,N-dimethylamino)[5-(4-acetylphenyl)-1-methyl-2-pyrrolyl]methylene]chromium(0) (3k). The general procedure A starting from **1b** (0.407 g, 1.0 mmol), BuLi (0.45 mL, 1.10 mmol), ZnBr_2 (0.225 g, 1.0 mmol), 4-iodoacetophenone (0.492 g, 2.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then DCM) afforded 0.320 g of the title compound (72%) as a yellow solid. Mp: >95 °C dec. ^1H NMR (CDCl_3): δ 2.62 (s, 3H, CH_3), 3.17 (s, 3H, CH_3), 3.35 (s, 3H, CH_3), 4.03 (s, 3H, CH_3), 5.90 (d, $J = 3.6$ Hz, 1H, CH), 6.35 (d, $J = 3.6$ Hz, 1H, CH), 7.45 (d, $J = 9.0$ Hz, 2H, CH), 7.98 (d, $J = 9.0$ Hz, 2H, CH). ^{13}C NMR (CDCl_3): δ 26.5, 32.4, 46.4, 50.5, 102.3, 110.8, 128.1, 128.6, 132.5, 135.1, 137.2, 145.1, 197.4, 216.7, 223.7, 272.6. IR (ATR): ν 2048 (m), 1891 (s), 1667 (m), 1601 (m), 1537 (w), 1444 (w), 1401 (w), 1361 (w), 1272 (w), 1183 (w), 1117 (w) cm^{-1} . Anal. Calcd for $\text{C}_{21}\text{H}_{18}\text{CrN}_2\text{O}_6$: C, 56.51; H, 4.06; N, 6.28. Found: C, 56.13; H, 4.14; N, 6.06.

Pentacarbonyl[(N,N-dimethylamino)[5-(2-pyridyl)-1-methyl-2-pyrrolyl]methylene]chromium(0) (3l). The general procedure A starting from **1b** (0.407 g, 1.0 mmol), BuLi (0.45 mL, 1.10 mmol), ZnBr_2 (0.225 g, 1.0 mmol), 2-bromopyridine (0.19 mL, 2.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then DCM) afforded 0.293 g of the title compound (72%) as a yellow solid. Mp: >89 °C dec. ^1H NMR (CDCl_3): δ 3.16 (s, 3H, CH_3), 3.66 (s, 3H, CH_3), 4.03 (s, 3H, CH_3), 5.91 (d, $J = 3.9$ Hz, 1H, CH), 6.59 (d, $J = 3.9$ Hz, 1H, CH), 7.05–7.09 (m, 1H, CH), 7.51–7.54 (m, 1H, CH), 7.61–7.67 (m, 1H, CH), 8.53–8.54 (m, 1H, CH). ^{13}C NMR (CDCl_3): δ 33.7, 46.4, 50.5, 102.4, 111.4, 120.4, 121.8, 131.3, 136.3, 145.8, 148.5, 152.5, 216.7, 223.8, 272.4. IR (ATR): ν 2049 (m), 1969 (w), 1880 (w), 1585 (w), 1532 (w), 1481 (w), 1445 (w), 1423 (w), 1399 (w), 1326 (w), 1227 (w), 1121 (w), 1071 (w) cm^{-1} . Anal. Calcd for $\text{C}_{18}\text{H}_{15}\text{CrN}_3\text{O}_5$: C, 53.34; H, 3.73; N, 10.37. Found: C, 53.33; H, 3.91; N, 10.34.

Pentacarbonyl[(N,N-dimethylamino)[5-(4-acetylphenyl)-2-furyl]methylene]chromium(0) (3m). The general procedure A starting from **1c** (0.300 g, 1.0 mmol), BuLi (0.41 mL, 0.99 mmol), ZnBr_2 (0.171 g, 1.0 mmol), 4-iodoacetophenone (0.374 g, 1.52 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.044 g, 0.038 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.210 g of the title compound (83%) as a yellow solid. Mp: >85 °C dec. ^1H NMR (CDCl_3): δ 2.61 (s, 3H, CH_3), 3.48 (s, 3H, CH_3), 4.01 (s, 3H, CH_3), 6.35 (d, $J = 3.3$ Hz, 1H, CH), 6.87 (d, $J = 3.3$ Hz, 1H, CH), 7.80–7.84 (m, 2H, CH), 7.99–8.02 (m, 2H, CH). ^{13}C NMR (CDCl_3): δ 26.5, 48.3, 51.7, 108.8, 109.4, 123.8, 129.0, 133.7, 136.1, 153.7, 158.8, 197.2, 217.0, 223.7, 258.5. Anal. Calcd for $\text{C}_{20}\text{H}_{15}\text{CrNO}_7$: C, 55.43; H, 3.49; N, 3.23. Found: C, 55.83; H, 3.75; N, 3.13.

Pentacarbonyl[(N,N-dimethylamino)[5-(4-formylphenyl)-2-furyl]methylene]tungsten(0) (3n). The general procedure A starting from **1c** (0.433 g, 1.1 mmol), BuLi (0.59 mL, 1.43 mmol), ZnBr_2 (0.247 g, 1.1 mmol), 4-iodobenzaldehyde (0.510 g, 2.2 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.064 g, 0.055 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.260 g of the title compound (56%) as a yellow solid. Mp: >115 °C dec. ^1H NMR (CDCl_3): δ 3.47 (s, 3H, CH_3), 4.01 (s, 3H, CH_3), 6.36 (d, $J = 3.3$ Hz, 1H, CH), 6.91 (d, $J = 3.3$ Hz, 1H, CH), 7.86–7.94 (m, 4H, CH), 10.01 (s, 1H, CH). ^{13}C NMR (acetone- d_6): δ 49.3, 52.8, 110.4, 111.0, 125.3, 131.3, 136.1, 136.9, 154.8, 160.7, 192.4, 218.4, 225.0, 253.4. Anal. Calcd for $\text{C}_{19}\text{H}_{13}\text{CrNO}_7$: C, 54.42; H, 3.12; N, 3.34. Found: C, 54.78; H, 3.22; N, 3.12.

Pentacarbonyl[(N,N-dimethylamino)[5-(4-acetylphenyl)-2-thienyl]methylene]tungsten(0) (3o). The general procedure A starting from **1d** (0.463 g, 1.0 mmol), BuLi (0.59 mL, 1.30 mmol), ZnBr_2 (0.225 g, 1.0 mmol), 4-iodoacetophenone (0.492 g, 2.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.480 g of the title compound (83%) as a yellow solid. Mp: >134 °C dec. ^1H NMR (CDCl_3): δ 2.60 (s, 3H, CH_3), 3.30 (s, 3H, CH_3), 3.94 (s, 3H, CH_3), 6.59 (d, $J = 3.6$ Hz, 1H, CH), 7.34 (d, $J = 3.6$ Hz, 1H, CH), 7.62–7.65 (m, 2H, CH), 7.94–7.97 (m, 2H, CH). ^{13}C NMR (CDCl_3): δ 26.5, 45.3, 53.8, 120.3, 124.3, 125.3, 129.1, 135.9, 137.8, 141.7, 154.3, 197.1, 198.1 ($J_{183\text{W}-13\text{C}} = 128$ Hz), 203.7, 249.5. IR (ATR): ν 2058 (m), 1970 (w), 1888 (w), 1674 (m), 1601 (w), 1537 (w), 1444 (w), 1395 (w), 1358 (w), 1258 (w), 1178 (w), 1107 (w), 1020 (w) cm^{-1} . Anal. Calcd for $\text{C}_{20}\text{H}_{15}\text{NO}_6\text{SW}$: C, 41.33; H, 2.60; N, 2.41; S, 5.52. Found: C, 41.28; H, 2.67; N, 2.31; S, 5.41.

Pentacarbonyl[(N,N-dimethylamino)[5-(2-pyridyl)-2-thienyl]methylene]tungsten(0) (3p). The general procedure A starting from **1d** (0.463 g, 1.0 mmol), BuLi (0.59 mL, 1.30 mmol), ZnBr_2 (0.225 g, 1.0 mmol), 2-bromopyridine (0.492 g, 2.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.420 g of the title compound (78%) as a yellow solid. Mp: >128 °C dec. ^1H NMR (CDCl_3): δ 3.28 (s, 3H, CH_3), 3.92 (s, 3H, CH_3), 6.58–6.61 (m, 1H, CH), 7.12–7.16 (m, 1H, CH), 7.45–7.47 (m, 1H, CH), 7.60–7.71 (m, 2H, CH), 8.52–8.54 (m, 1H, CH). ^{13}C NMR (CDCl_3): δ 45.2, 53.8, 118.5, 120.2, 122.1, 124.1, 136.6, 143.7, 149.5, 151.9, 155.3, 198.1 ($J_{183\text{W}-13\text{C}} = 128$ Hz), 203.8, 250.3. IR (ATR): ν 2060 (m), 1978 (w), 1883 (s), 1582 (w), 1533 (w), 1469 (w), 1430 (w), 1400 (w), 1293 (w), 1218 (w), 1154 (w), 1106 (w) cm^{-1} . Anal. Calcd for $\text{C}_{17}\text{H}_{12}\text{N}_2\text{O}_5\text{SW}$: C, 37.80; H, 2.24; N, 5.19; S, 5.94. Found: C, 37.82; H, 2.30; N, 5.14; S, 5.95.

Pentacarbonyl[(N,N-dimethylamino)(4'-formyl-4-biphenyl)methylene]chromium(0) (5a). The general procedure A starting from **4a** (0.202 g, 0.50 mmol), BuLi (0.27 mL, 0.65 mmol), ZnBr_2 (0.113 g, 0.50 mmol), a solution of 4-iodobenzaldehyde (0.232 g, 1.0 mmol) in dry THF (1 mL) and $\text{Pd}(\text{PPh}_3)_4$ (0.029 g, 0.025 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.140 g of the title compound (65%) as a yellow solid. Mp: >110 °C dec. ^1H NMR (CDCl_3): δ 3.13 (s, 3H, CH_3), 4.04 (s, 3H, CH_3), 6.83 (d, $J = 8.4$ Hz, 2H, CH), 7.69 (d, $J = 8.4$ Hz, 2H, CH), 7.78 (d, $J = 8.4$ Hz, 2H, CH), 7.96 (d, $J = 8.4$ Hz, 2H, CH), 10.06 (s, 1H, CH). ^{13}C NMR (CDCl_3): δ 46.1, 51.3, 119.7, 127.3, 127.5, 130.2, 135.2, 136.9, 146.1, 152.3, 191.8, 217.0, 223.6, 274.2. IR (ATR): ν 2052 (m), 1973 (w), 1871 (s), 1693 (m), 1604 (m), 1543 (m), 1488 (m), 1401 (m), 1212 (w), 1172 (w), 1118 (w) cm^{-1} . Anal. Calcd for $\text{C}_{21}\text{H}_{15}\text{CrNO}_6$: C, 58.75; H, 3.52; N, 3.26. Found: C, 59.25; H, 3.53; N, 2.83.

Pentacarbonyl[(N,N-dimethylamino)(4'-acetyl-4-biphenyl)methylene]chromium(0) (5b). The general procedure A starting from **4a** (0.202 g, 0.50 mmol), BuLi (0.27 mL, 0.65 mmol), ZnBr_2 (0.113 g, 0.50 mmol), 4-iodoacetophenone (0.246 g, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.029 g, 0.025 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.170 g of the title compound (77%) as a yellow solid. Mp: >129 °C dec. ^1H NMR (CDCl_3): δ 2.63 (s, 3H, CH_3), 3.11 (s, 3H, CH_3), 4.01 (s, 3H, CH_3), 6.82 (d, $J = 7.8$ Hz, 2H, CH), 7.65–7.71 (m, 4H, CH), 8.02 (d, $J = 7.8$

Hz, 2H, CH). ^{13}C NMR (CDCl_3): δ 26.6, 46.0, 51.3, 119.6, 126.8, 127.3, 128.8, 135.8, 137.0, 144.6, 152.2, 197.6, 217.0, 223.7, 274.0. IR (ATR): ν 2054 (m), 1985 (m), 1868 (s), 1674 (s), 1601 (m), 1536 (m), 1400 (m), 1351 (m), 1274 (m), 1115 (w) cm^{-1} . Anal. Calcd for $\text{C}_{22}\text{H}_{17}\text{CrNO}_6$: C, 59.60; H, 3.86; N, 3.16. Found: C, 59.35; H, 3.78; N, 3.06.

Pentacarbonyl[(*N,N*-dimethylamino)(4'-ethoxycarbonyl-4-biphenyl)methylene]chromium(0) (5c). The general procedure A starting from **4a** (0.202 g, 0.50 mmol), BuLi (0.27 mL, 0.65 mmol), ZnBr_2 (0.113 g, 0.50 mmol), ethyl 4-iodobenzoate (0.17 mL, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.029 g, 0.025 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.170 g of the title compound (72%) as a yellow solid. Alternatively, the title compound was prepared following the general procedure B starting from **4a** (0.335 g, 0.83 mmol), $i\text{PrMgCl}\cdot\text{LiCl}$ (0.85 mL, 1.05 mmol), ZnBr_2 (0.225 g, 1.0 mmol), ethyl 4-iodobenzoate (0.276 g, 1.0 mmol) and $\text{PdCl}_2\text{XantPhos}$ (0.031 g, 0.042 mmol) giving 0.290 g of the **5c** (74%). Mp: $>117^\circ\text{C}$ dec. ^1H NMR (CDCl_3): δ 1.42 (t, J = 6.9 Hz, 3H, CH_3), 3.12 (s, 3H, CH_3), 4.03 (s, 3H, CH_3), 4.41 (q, J = 6.9 Hz, 2H, CH_2), 6.81 (d, J = 8.1 Hz, 2H, CH), 7.65–7.69 (m, 4H, CH), 8.11 (d, J = 8.1 Hz, 2H, CH). ^{13}C NMR (CDCl_3): δ 14.3, 46.0, 51.3, 60.9, 119.6, 126.6, 127.4, 129.3, 130.0, 137.2, 144.4, 152.1, 166.4, 217.1, 223.7, 274.4. IR (ATR): ν 2052 (m), 1970 (w), 1937 (m), 1878 (s), 1710 (m), 1608 (m), 1541 (m), 1402 (w), 1365 (w), 1274 (m), 1103 (m) cm^{-1} . Anal. Calcd for $\text{C}_{23}\text{H}_{19}\text{CrNO}_7$: C, 58.35; H, 4.05; N, 2.96. Found: C, 58.64; H, 4.23; N, 2.83.

Pentacarbonyl[(*Z*)-{(*N,N*-dimethylamino)[4-(2-ethoxycarbonylvinyl)phenyl]methylene}chromium(0) (5d). The general procedure A starting from **4a** (0.202 g, 0.50 mmol), BuLi (0.27 mL, 0.65 mmol), ZnBr_2 (0.113 g, 0.50 mmol), (*Z*)-ethyl 3-iodopropenoate (0.226 g, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.029 g, 0.025 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2) afforded 0.150 g of the title compound (71%) as a yellow solid. Mp: $77\text{--}80^\circ\text{C}$. ^1H NMR (CDCl_3): δ 1.27 (t, J = 6.3 Hz, 3H, CH_3), 3.08 (s, 3H, CH_3), 4.00 (s, 3H, CH_3), 4.19 (q, J = 6.3 Hz), 5.94 (d, J = 12.9 Hz, 1H, CH), 6.70 (d, J = 7.8 Hz, 2H, CH), 6.88 (d, J = 12.9 Hz, 1H, CH), 7.69 (d, J = 7.8 Hz, 2H, CH). ^{13}C NMR (CDCl_3): δ = 14.0, 46.1, 51.3, 60.3, 118.6, 119.7, 130.5, 132.3, 142.1, 152.8, 166.2, 217.0, 223.6, 274.5. IR (ATR): ν 2052 (m), 1969 (w), 1939 (m), 1871 (s), 1706 (m), 1619 (m), 1542 (m), 1502 (w), 1478 (w), 1439 (w), 1403 (m), 1178 (m) cm^{-1} . Anal. Calcd for $\text{C}_{19}\text{H}_{17}\text{CrNO}_7$: C, 53.91; H, 4.05; N, 3.31. Found: C, 53.52; H, 4.06; N, 3.10.

Pentacarbonyl[(*N,N*-dimethylamino)[4-(2-pyridyl)phenyl]methylene]chromium(0) (5e). The general procedure A starting from **4a** (0.202 g, 0.50 mmol), BuLi (0.27 mL, 0.65 mmol), ZnBr_2 (0.113 g, 0.50 mmol), 2-bromopyridine (0.095 mL, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.029 g, 0.025 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then DCM) afforded the title compound 0.170 g (85%) as a yellow solid. Alternatively, the title compound was prepared following the general procedure B starting from **4a** (0.335 g, 0.83 mmol), BuLi (0.43 mL, 1.05 mmol), ZnBr_2 (0.225 g, 1.0 mmol), 2-bromopyridine (0.158 g, 1.0 mmol) and $\text{PdCl}_2\text{XantPhos}$ (0.031 g, 0.042 mmol) giving 0.173 g of the **5e** (52%). Mp: $>123^\circ\text{C}$ dec. ^1H NMR (CDCl_3): δ 3.06 (s, 3H, CH_3), 3.99 (s, 3H, CH_3), 6.82 (d, J = 7.8 Hz, 2H, CH), 7.20 (m, 1H, CH), 7.73 (m, 2H, CH), 8.03 (d, J = 7.8 Hz, 2H, CH), 8.67 (m, 1H, CH). ^{13}C NMR (CDCl_3): δ 46.0, 51.3, 119.4, 120.2, 122.2, 127.1, 136.8 (2C), 149.6, 152.8, 156.4, 217.0, 223.7, 274.3. IR (ATR): ν 2050 (m), 1972 (m), 1932 (w), 1903 (s), 1868 (s), 1585 (m), 1571 (m), 1464 (m), 1431 (m), 1401 (m), 1233 (w), 1121 (w) cm^{-1} . Anal. Calcd for $\text{C}_{19}\text{H}_{14}\text{CrN}_2\text{O}_5$: C, 56.72; H, 3.51; N, 6.96. Found: C, 56.93; H, 3.58; N, 6.69.

Pentacarbonyl[(*N,N*-dimethylamino)(4'-acetyl-4-biphenyl)methylene]tungsten(0) (5f). The general procedure A starting from **4b** (0.536 g, 1.0 mmol), BuLi (0.40 mL, 1.30 mmol), ZnBr_2 (0.225 g, 1.0 mmol), 4-iodoacetophenone (0.492 g, 2.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.050 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then DCM) afforded 0.300 g of the title compound (52%) as a yellow solid. Mp: $>150^\circ\text{C}$ dec. ^1H NMR (CDCl_3): δ 2.64 (s, 3H, CH_3), 3.12 (s, 3H, CH_3), 3.96 (s, 3H, CH_3), 6.88 (d, J = 8.4

Hz, 2H, CH), 7.67–7.72 (2, 4H, CH), 8.04 (d, J = 8.4 Hz, 2H, CH). ^{13}C NMR (CDCl_3): δ 26.6, 44.5, 53.7, 112.0, 126.9, 127.2, 128.9, 135.9, 137.4, 144.6, 152.6, 197.7, 198.3 ($J_{183\text{W}-13\text{C}}$ = 128 Hz), 203.9, 253.7. IR (ATR): ν 2060 (m), 1874 (s), 1675 (m), 1604 (w), 1544 (w), 1402 (w), 1355 (w), 1267 (w), 1122 (w), 1001 (w) cm^{-1} . Anal. Calcd for $\text{C}_{22}\text{H}_{17}\text{NO}_6\text{W}$: C, 45.94; H, 2.98; N, 2.44. Found: C, 45.77; H, 3.06; N, 2.33.

Pentacarbonyl[(*N,N*-dimethylamino)[4-(2-pyridyl)phenyl]methylene]tungsten(0) (5g). The general procedure A starting from **4b** (0.536 g, 1.0 mmol), BuLi (0.40 mL, 1.30 mmol), ZnBr_2 (0.225 g, 1.0 mmol), 2-bromopyridine (0.19 mL, 2.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.050 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then DCM) afforded 0.320 g of the title compound (60%) as a yellow solid. Mp: $>142^\circ\text{C}$ dec. ^1H NMR (CDCl_3): δ 3.07 (s, 3H, CH_3), 3.94 (s, 3H, CH_3), 6.86–6.90 (m, 2H, CH), 7.20–7.26 (m, 1H, CH), 7.73–7.76 (m, 2H, CH), 8.03–8.06 (m, 2H, CH), 8.66–8.69 (m, 1H, CH). ^{13}C NMR (CDCl_3): δ 44.4, 53.7, 119.8, 120.3, 122.2, 127.0, 136.8, 137.2, 149.6, 153.2, 156.4, 198.3 ($J_{183\text{W}-13\text{C}}$ = 128 Hz), 204.0, 256.9. IR (ATR): ν 2059 (m), 1970 (m), 1866 (s), 1545 (w), 1464 (w), 1431 (w), 1402 (w), 1122 (w), 1013 (w) cm^{-1} . Anal. Calcd for $\text{C}_{19}\text{H}_{14}\text{N}_2\text{O}_5\text{W}$: C, 42.72; H, 2.64; N, 5.24. Found: C, 43.08; H, 2.80; N, 5.07.

Pentacarbonyl[ethoxy(5-bromo-2-thienyl)methylene]chromium(0) (6a). Butyllithium (3.13 mL, 5.0 mmol, 1.6 M solution in hexanes) was added to a solution of 2,5-dibromothiophene (1.21 g, 5.0 mmol) in dry ether (20 mL) cooled to -78°C . The resultant mixture was stirred for 30 min at -78°C , then $\text{Cr}(\text{CO})_6$ (1.1 g, 5 mmol) was added and the mixture was stirred for 2 h at ambient temperature. The solvents were removed under reduced pressure, and the isolated oily residue was dissolved in water (20 mL) and cooled to 0°C with ice–water bath. Then Et_3OBF_4 (1.43 g, 7.5 mmol) was added, and the reaction mixture was stirred for 10 min at 0°C and precipitated red solid was isolated by extraction with ethyl acetate (3×100 mL). The combined organic layers were dried (MgSO_4) and concentrated under reduced pressure, and column chromatography (silica gel, hexane) gave 1.60 g of the title compound (78%) as a red solid. Mp: $64\text{--}66^\circ\text{C}$. ^1H NMR (CDCl_3): δ 1.65 (t, J = 7.2 Hz, 3H, CH_3), 5.14 (q, J = 7.2 Hz, 2H, CH_2), 7.23 (d, J = 4.5 Hz, 1H, CH), 7.97 (d, J = 4.5 Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.1, 76.0, 126.6, 132.2, 141.1, 155.5, 216.8, 223.0, 314.0. IR (ATR): ν 2056 (m), 1987 (w), 1900 (s), 1399 (m), 1358 (m), 1316 (m), 1181 (m), 1108 (w), 1069 (w) cm^{-1} . Anal. Calcd for $\text{C}_{12}\text{H}_7\text{BrCrO}_5$: C, 35.06; H, 1.72; S, 7.80. Found: C, 34.99; H, 1.73; S, 7.76.

Pentacarbonyl[ethoxy(5-bromo-2-furyl)methylene]chromium(0) (6b). This compound was prepared according to the modified literature procedure:¹⁹ BuLi (8.64 mL, 21.0 mmol) was added to the solution of furane (1.45 mL, 20.0 mmol) in dry THF (40 mL) cooled to -78°C . The resultant mixture was stirred for 1 h at 0°C , then $\text{Cr}(\text{CO})_6$ (2.20 g, 10 mmol) was added, and the mixture was stirred for another 2 h at ambient temperature. The reaction mixture was then cooled to -78°C and BuLi (10.70 mL, 26 mmol) was added. After stirring for 1 h at -78°C , $\text{C}_2\text{Br}_2\text{F}_4$ (4.8 mL, 40 mmol) was added and the mixture was allowed to warm to ambient temperature. After 1 h, the solvents were removed under reduced pressure, and the isolated oily residue was dissolved in water (20 mL) and cooled to 0°C with ice–water bath. Then Et_3OBF_4 (5.70 g, 30.0 mmol) was added, the reaction mixture was stirred for 10 min at 0°C and precipitated red solid was isolated by extraction with ethyl acetate (3×100 mL). The combined organic layers were dried (MgSO_4), concentrated under reduced pressure and column chromatography (silica gel, hexane then hexane/DCM 9/1) gave 3.50 g of the title compound (44%) as a red solid. Mp: $>59^\circ\text{C}$ dec. ^1H NMR (CDCl_3): δ 1.65 (t, J = 6.9 Hz, 3H, CH_3), 5.14 (q, J = 6.9 Hz, 2H, CH_2), 6.54 (m, 1H, CH), 6.95 (d, J = 3.9 Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.2, 75.8, 115.2 (2C), 132.5, 164.9, 216.7, 223.8, 307.9. IR (ATR): ν 2059 (m), 1981 (w), 1899 (s), 1542 (w), 1418 (w), 1357 (w), 1205 (m), 1014 (w) cm^{-1} . Anal. Calcd for $\text{C}_{12}\text{H}_7\text{BrCrO}_5$: C, 36.48; H, 1.79. Found: C, 36.63; H, 1.87.

Pentacarbonyl[ethoxy(5-bromo-2-thienyl)methylene]tungsten(0) (6c). Butyllithium (6.25 mL, 10.0 mmol) was added to a

solution of 2,5-dibromothiophene (2.42 g, 10.0 mmol) in dry ether (40 mL) cooled to -78°C . The resultant mixture was stirred for 30 min at -78°C , then $\text{W}(\text{CO})_6$ (3.52 g, 10 mmol) was added and the mixture was stirred for another 2 h at ambient temperature. The solvents were removed under reduced pressure, and the isolated oily residue was dissolved in water (20 mL) and cooled to 0°C with ice–water bath. Then Et_3OBF_4 (3.23 g, 17.0 mmol) was added, the reaction mixture was stirred for 10 min at 0°C , and precipitated red solid was isolated by extraction with ethyl acetate (3×100 mL). Combined organic layers were dried (MgSO_4) and concentrated under reduced pressure, and column chromatography (silica gel, hexane) gave 4.15 g of the title compound (76%) as a red solid. Mp: $58\text{--}61^{\circ}\text{C}$. ^1H NMR (CDCl_3): δ 1.64 (t, $J = 7.2$ Hz, 3H, CH_3), 4.96 (q, $J = 7.2$ Hz, 2H, CH_2), 7.22 (d, $J = 4.2$ Hz, 1H, CH), 7.88 (d, $J = 4.2$ Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 14.9, 78.5, 127.2, 132.2, 142.1, 158.7, 197.3 ($J_{183\text{W--}^{13}\text{C}} = 127$ Hz), 202.2, 288.2. IR (ATR): ν 2065 (m), 1883 (s), 1506 (w), 1394 (m), 1314 (w), 1216 (w), 1163 (m), 1060 (w), 1015 (w) cm^{-1} . Anal. Calcd for $\text{C}_{12}\text{H}_7\text{BrO}_6\text{SW}$: C, 26.54; H, 1.30; S, 5.91. Found: C, 26.64; H, 1.35; S, 6.00.

Pentacarbonyl[ethoxy(5-bromo-1-methyl-2-pyrrolyl)methylene]chromium(0) (6d). Butyllithium (17.60 mL, 38.64 mmol) was added to a solution of 2,5-dibromo-1-methylpyrrol (9.23 g, 38.64 mmol) in dry THF (150 mL) cooled to -78°C . The resultant mixture was stirred for 30 min at -78°C , then $\text{Cr}(\text{CO})_6$ (8.50 g, 38.64 mmol) was added and the mixture was stirred for another 2 h at ambient temperature. The solvents were then removed under reduced pressure, and the isolated oily residue was dissolved in water (20 mL) and cooled to 0°C with ice–water bath. Then Et_3OBF_4 (11.0 g, 58.0 mmol) was added, the reaction mixture was stirred for 10 min at 0°C and precipitated red solid was isolated by extraction with ethyl acetate (3×100 mL). Combined organic layers were dried (MgSO_4), concentrated under reduced pressure and column chromatography (silica gel, hexane then hexan/DCM 4/1) gave 6.80 g of the title compound (43%) as a red solid. Mp: $>65^{\circ}\text{C}$ dec. ^1H NMR (CDCl_3): δ 1.68 (t, $J = 7.2$ Hz, 3H, CH_3), 3.78 (s, 1H, CH_3), 5.11 (q, $J = 7.2$ Hz, 2H, CH_2), 6.42 (m, 1H, CH), 7.67 (d, $J = 4.5$ Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.4, 38.2, 75.8, 114.5, 121.0, 133.2, 145.3, 217.6, 222.9, 302.4. IR (ATR): ν 2052 (m), 1895 (s), 1501 (w), 1468 (w), 1446 (w), 1419 (w), 1392 (w), 1361 (w), 1324 (w), 1298 (w), 1229 (w), 1194 (w), 1135 (w), 1102 (w), 1081 (w), 1043 (w), 1003 (w) cm^{-1} . Anal. Calcd for $\text{C}_{13}\text{H}_{10}\text{BrCrNO}_6$: C, 38.26; H, 2.47; N, 3.43. Found: C, 38.53; H, 2.72; N, 3.56.

Pentacarbonyl[ethoxy(5-bromo-2-furyl)methylene]molybdenum(0) (10). Butyllithium (5.0 mL, 12.1 mmol) was added to a solution of 2,5-dibromothiophene (2.80 g, 11.6 mmol) in dry THF (40 mL) cooled to -78°C . The resultant mixture was stirred for 30 min at -78°C , then $\text{Mo}(\text{CO})_6$ (3.062 g, 11.6 mmol) was added and the mixture was stirred for another 2 h at ambient temperature. The solvents were then removed under reduced pressure, and the isolated oily residue was dissolved in water (20 mL) and cooled to 0°C with ice–water bath. Then Et_3OBF_4 (3.31 g, 17.4 mmol) was added, the reaction mixture was stirred for 10 min at 0°C and precipitated red solid was isolated by extraction with ethyl acetate (3×100 mL). Combined organic layers were dried (MgSO_4) and concentrated under reduced pressure, and column chromatography (silica gel, hexane) gave 2.69 g of the title compound (51%) as a red solid. Mp: $53\text{--}55^{\circ}\text{C}$. ^1H NMR (CDCl_3): δ 1.64 (t, $J = 6.9$ Hz, 3H, CH_3), 5.05 (q, $J = 6.9$ Hz, 2H, CH_2), 7.34 (d, $J = 4.5$ Hz, 1H, CH), 7.94 (d, $J = 4.5$ Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.0, 77.7, 127.4, 132.2, 156.4, 205.8, 212.5, 304.6. Anal. Calcd for $\text{C}_{12}\text{H}_7\text{BrMoO}_6\text{S}$: C, 31.67; H, 1.55; S, 7.05. Found: C, 32.12; H, 1.67; S, 7.47.

Pentacarbonyl[ethoxy(5-butyl-2-thienyl)methylene]chromium(0) (7a). The general procedure B starting from BuLi (0.21 mL, 0.50 mmol), ZnBr_2 (0.113 g, 0.50 mmol), **6a** (0.247 g, 0.60 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.029 g, 0.025 mmol) followed by column chromatography (silica gel, hexane) afforded 0.130 g of the title compound (67%) as a red oil. ^1H NMR (CDCl_3): δ 0.96 (t, $J = 7.2$ Hz, 3H, CH_3), 1.38–1.46 (m, 2H, CH_2), 1.63–1.73 (m, 5H, CH_2 and CH_3), 2.84 (t, $J = 7.8$ Hz, 2H, CH_2), 5.18 (q, $J = 6.9$ Hz, 2H, CH_2), 6.95 (d, $J = 4.0$ Hz, 1H, CH), 8.13 (d, $J = 4.0$ Hz, 1H, CH). ^{13}C NMR

(CDCl_3): δ 13.7, 15.1, 22.2, 30.4, 33.2, 75.6, 127.1, 142.5, 152.7, 158.3, 217.3, 223.2, 312.4. IR (ATR): ν 2054 (m), 1901 (s), 1431 (w), 1238 (w), 1189 (w) cm^{-1} . Anal. Calcd for $\text{C}_{16}\text{H}_{16}\text{CrO}_6\text{S}$: C, 49.48; H, 4.15; S, 8.26. Found: C, 49.70; H, 3.90; S, 8.06.

Pentacarbonyl[ethoxy(5-phenyl-2-thienyl)methylene]chromium(0) (7b). The general procedure B starting from BuLi (0.40 mL, 0.99 mmol), bromobenzene (0.095 mL, 0.90 mmol), ZnBr_2 (0.203 g, 0.90 mmol), **6a** (0.411 g, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.052 g, 0.045 mmol) followed by column chromatography (silica gel, hexane) afforded 0.330 g of the title compound (90%) as a red solid. Mp: $104\text{--}105^{\circ}\text{C}$. ^1H NMR (CDCl_3): δ 1.70 (t, $J = 7.2$ Hz, 3H, CH_3), 5.19 (q, $J = 7.2$ Hz, 2H, CH_2), 7.42–7.47 (m, 4H, CH), 7.70–7.73 (m, 2H, CH), 8.27 (d, $J = 4.5$ Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.2, 75.8, 125.0, 126.4, 129.1, 129.7, 132.9, 142.6, 153.3, 153.9, 217.2, 223.3, 313.1. IR (ATR): ν 2050 (m), 1966 (w), 1894 (s), 1430 (m), 1346 (m), 1251 (m), 1179 (m), 1066 (w), 1011 (w) cm^{-1} . Anal. Calcd for $\text{C}_{18}\text{H}_{12}\text{CrO}_6\text{S}$: C, 52.94; H, 2.96; S, 7.85. Found: C, 52.70; H, 2.85; S, 7.67.

Pentacarbonyl[ethoxy(5-(6-methoxy-2-naphthyl)-2-thienyl)methylene]chromium(0) (7c). The general procedure B starting from BuLi (0.40 mL, 0.99 mmol), 2-methoxy-6-bromonaphthalene (0.213 g, 0.90 mmol), ZnBr_2 (0.203 g, 0.90 mmol), **6a** (0.411 g, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.052 g, 0.045 mmol) followed by column chromatography (silica gel, hexane) afforded 0.440 g of the title compound (100%) as a red solid. Mp: $>131^{\circ}\text{C}$ dec. ^1H NMR (CDCl_3): δ 1.70 (t, $J = 7.2$ Hz, 3H, CH_3), 3.95 (s, 3H, CH_3), 5.18 (q, $J = 7.2$ Hz, 2H, CH_2), 7.14–7.25 (m, 2H, CH), 7.55 (d, $J = 4.5$ Hz, 1H, CH), 7.75 (m, 3H, CH), 8.12 (s, 1H, CH), 8.29 (d, $J = 4.5$ Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.22, 55.3, 75.6, 105.8, 119.8, 124.3, 124.8, 125.5, 127.6, 128.0, 128.7, 129.9, 135.2, 143.0, 152.7, 154.5, 158.7, 217.2, 223.4, 311.7. IR (ATR): ν 2051 (m), 1899 (s), 1619 (m), 1479 (m), 1425 (m), 1382 (m), 1310 (w), 1234 (w), 1176 (m), 1068 (w) cm^{-1} . Anal. Calcd for $\text{C}_{23}\text{H}_{16}\text{CrO}_6\text{S}$: C, 56.56; H, 3.30; S, 6.56. Found: C, 57.18; H, 3.20; S, 6.12.

Pentacarbonyl[ethoxy(5-(5-chloro-2-thienyl)-2-thienyl)methylene]chromium(0) (7d). The general procedure B starting from BuLi (0.40 mL, 0.99 mmol), 2-chlorothiophene (0.083 mL, 0.90 mmol), ZnBr_2 (0.203 g, 0.90 mmol), **6a** (0.411 g, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.052 g, 0.045 mmol) followed by column chromatography (silica gel, hexane) afforded 0.300 g of the title compound (74%) as a red solid. Mp: $>140^{\circ}\text{C}$ dec. ^1H NMR (CDCl_3): δ 1.67 (t, $J = 6.9$ Hz, 3H, CH_3), 5.15 (q, $J = 6.9$ Hz, 2H, CH_2), 6.90 (d, $J = 4.2$ Hz, 1H, CH), 7.18 (d, $J = 3.9$ Hz, 1H, CH), 7.22 (d, $J = 3.9$ Hz, 1H, CH), 8.16 (d, $J = 4.2$ Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.2, 75.8, 125.1, 125.4, 127.6, 132.1, 134.7, 142.6, 145.5, 152.3, 217.1, 223.3, 312.5. IR (ATR): ν 2052 (m), 1893 (s), 1426 (m), 1354 (m), 1289 (m), 1235 (m), 1170 (m), 1103 (m), 1058 (m) cm^{-1} . Anal. Calcd for $\text{C}_{16}\text{H}_9\text{ClCrO}_6\text{S}$: C, 42.82; H, 2.02; S, 14.29. Found: C, 42.43; H, 2.06; S, 13.97.

Pentacarbonyl[ethoxy(5-(2-phenylethynyl)-2-thienyl)methylene]chromium(0) (7e). The general procedure B starting from BuLi (0.40 mL, 0.99 mmol), phenylacetylene (0.099 mL, 0.90 mmol), ZnBr_2 (0.203 g, 0.90 mmol), **6a** (0.411 g, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.052 g, 0.045 mmol) followed by column chromatography (silica gel, hexane) afforded 0.240 g of the title compound (62%) as a red oil. ^1H NMR (CDCl_3): δ 1.68 (t, $J = 7.2$ Hz, 3H, CH_3), 5.17 (q, $J = 7.2$ Hz, 2H, CH_2), 7.33 (d, $J = 4.5$ Hz, 1H, CH), 7.38–7.40 (m, 3H, CH), 7.53–7.56 (m, 2H, CH), 8.15 (d, $J = 4.5$ Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.1, 75.9, 82.4, 99.1, 122.0, 128.5, 129.3, 131.7, 132.3, 133.1, 140.6, 154.5, 216.9, 223.3, 314.4. IR (ATR): ν 2052 (s), 1894 (s), 1487 (m), 1425 (m), 1267 (w), 1211 (m), 1186 (m), 1114 (w), 1050 (w) cm^{-1} . Anal. Calcd for $\text{C}_{20}\text{H}_{12}\text{CrO}_6\text{S}$: C, 55.56; H, 2.80; S, 7.42. Found: C, 55.22; H, 2.76; S, 7.27.

Pentacarbonyl[ethoxy(5-(2-methylphenyl)-2-thienyl)methylene]chromium(0) (7f). The reaction mixture prepared according the general procedure B from BuLi (0.40 mL, 0.99 mmol), 2-iodotoluene (0.196 mL, 0.90 mmol), ZnBr_2 (0.203 g, 0.90 mmol), **6a** (0.411 g, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.052 g, 0.045 mmol) was stirred for 20 h at ambient temperature. Subsequent column chromatography (silica gel, hexane) afforded 0.160 g of the title

compound (42%) as a red solid. Mp: 104–106 °C. ^1H NMR (CDCl_3): δ 1.70 (t, J = 7.2 Hz, 3H, CH_3), 2.50 (s, 3H, CH_3), 5.21 (q, J = 7.2 Hz, 2H, CH_2), 7.24–7.34 (m, 4H, CH), 7.45 (m, 1H, CH), 8.29 (d, J = 3.9 Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.2, 21.2, 75.8, 126.2, 128.6, 129.2, 130.0, 131.1, 132.8, 136.1, 141.5, 153.5, 154.1, 217.2, 223.3, 313.8. IR (ATR): ν 2055 (m), 1986 (w), 1887 (s), 1430 (m), 1358 (w), 1340 (m), 1207 (m), 1180 (m), 1072 (m), 1010 (m) cm^{-1} . Anal. Calcd for $\text{C}_{19}\text{H}_{14}\text{CrO}_6\text{S}$: C, 54.03; H, 3.34; S, 7.59. Found: C, 54.15; H, 3.23; S, 7.44.

Pentacarbonyl[ethoxy(5-vinyl-2-thienyl)methylene]chromium(0) (7g). The general procedure B starting from vinylmagnesium chloride (0.57 mL, 0.90 mmol), ZnBr_2 (0.203 g, 0.90 mmol), **6a** (0.411 g, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.052 g, 0.045 mmol) followed by column chromatography (silica gel, hexane) afforded 0.220 g of the title compound (68%) as a red oil. ^1H NMR (CDCl_3): δ 1.66 (t, J = 7.2 Hz, 3H, CH_3), 5.16 (q, J = 7.2 Hz, 2H, CH_2), 5.41 (d, J = 10.8 Hz, 1H, CH), 5.88 (d, J = 17.1 Hz, 1H, CH), 6.78 (dd, J = 10.8, 17.4 Hz, 1H, CH), 7.12 (d, J = 4.5 Hz, 1H, CH), 8.15 (d, J = 4.5 Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.2, 75.8, 118.4, 127.4, 129.5, 142.1, 151.7, 152.7, 217.1, 223.3, 313.8. IR (ATR): ν 2054 (w), 1890 (s), 1433 (w), 1181 (w), 1057 (w), 1010 (w) cm^{-1} . Anal. Calcd for $\text{C}_{14}\text{H}_{10}\text{CrO}_6\text{S}$: C, 46.93; H, 2.81; S, 8.95. Found: C, 46.59; H, 2.80; S, 8.62.

Pentacarbonyl[ethoxy[4-(ethoxycarbonylphenyl)-2-thienyl]methylene]chromium(0) (7h). The general procedure B starting from $i\text{PrMgCl}\cdot\text{LiCl}$ (0.77 mL, 0.95 mmol), ethyl 4-iodobenzoate (0.248 g, 0.90 mmol), ZnBr_2 (0.203 g, 0.90 mmol), **6a** (0.411 g, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.052 g, 0.045 mmol) followed by column chromatography (silica gel, hexane/DCM, 4/1 then 1/1) afforded 0.220 g of the title compound (51%) as a red solid. Mp: >123 °C dec. ^1H NMR (CDCl_3): δ 1.41 (t, J = 6.9 Hz, 3H, CH_3), 1.70 (t, J = 6.9 Hz, 3H, CH_3), 4.40 (q, J = 6.9 Hz, 2H, CH_2), 5.19 (q, J = 6.9 Hz, 3H, CH_3), 7.53 (d, J = 4.5 Hz, 1H, CH), 7.75 (d, J = 8.4 Hz, 2H, CH), 8.09 (d, J = 8.4 Hz, 2H, CH), 8.25 (d, J = 4.2 Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 14.3, 15.2, 61.2, 75.9, 125.95, 126.04, 130.3, 131.0, 137.0, 142.1, 151.6, 154.2, 165.8, 217.0, 223.3, 314.4. IR (ATR): ν 2052 (m), 1897 (s), 1703 (m), 1604 (w), 1425 (w), 1361 (w), 1271 (w), 1245 (w), 1174 (w), 1105 (w), 1011 (w) cm^{-1} . Anal. Calcd for $\text{C}_{21}\text{H}_{16}\text{CrO}_8\text{S}$: C, 52.50; H, 3.36; S, 6.67. Found: C, 52.38; H, 3.25; S, 6.74.

Pentacarbonyl[ethoxy[5-(4'-carbonitrile-4-biphenyl)-2-thienyl]methylene]chromium(0) (7i). The general procedure B starting from $i\text{PrMgCl}\cdot\text{LiCl}$ (0.77 mL, 0.95 mmol), 4'-iodobiphenyl-4-carbonitrile (0.275 g, 0.90 mmol), ZnBr_2 (0.203 g, 0.90 mmol), **6a** (0.411 g, 1.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.052 g, 0.045 mmol) followed by column chromatography (silica gel, hexane/DCM 4/1 then 1/1) afforded 0.350 g of the title compound (76%) as a red solid. Mp: >155 °C dec. ^1H NMR (CDCl_3): δ 1.70 (t, J = 7.2 Hz, 3H, CH_3), 5.19 (q, J = 7.2 Hz, 2H, CH_2), 7.52 (d, J = 4.5 Hz, 1H, CH), 7.65–7.83 (m, 8H, CH), 8.26 (d, J = 4.5 Hz, 1H). ^{13}C NMR (125.8 MHz, $\text{THF}-d_6$): δ 15.3, 77.3, 112.6, 118.9, 126.6, 127.7, 128.3, 128.7, 133.4, 134.0, 141.1, 144.0, 144.8, 154.0, 154.4, 218.1, 224.4, 313.6. IR (ATR): ν 2225 (w), 1893 (s), 1606 (m), 1431 (m), 1353 (w), 1252 (w), 1174 (m), 1068 (m), 1003 (w) cm^{-1} . Anal. Calcd for $\text{C}_{25}\text{H}_{15}\text{CrO}_6\text{S}$: C, 58.94; H, 2.97; N, 2.75; S, 6.29. Found: C, 58.67; H, 2.86; N, 2.62; S, 6.18.

Pentacarbonyl[ethoxy(5-phenyl-2-furyl)methylene]chromium(0) (7j). The general procedure B starting from phenylmagnesium chloride (0.60 mL, 1.0 mmol), ZnBr_2 (0.225 g, 1.0 mmol), **6b** (0.435 g, 1.1 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, hexane then hexane/DCM 9/1) afforded 0.370 g of the title compound (94%) as a red solid. Mp: >131 °C dec. ^1H NMR (CDCl_3): δ 1.65 (t, J = 7.2 Hz, 3H, CH_3), 5.14 (q, J = 7.2 Hz, 2H, CH_2), 6.85 (d, J = 3.6 Hz, 1H, CH), 7.03 (d, J = 3.6 Hz, 1H, CH), 7.42–7.52 (m, 3H, CH), 7.90–7.94 (m, 2H, CH). ^{13}C NMR (CDCl_3): δ 15.3, 75.4, 108.4, 113.2, 125.1, 129.07, 129.12, 129.9, 162.2, 163.3, 217.2, 224.1, 306.6. IR (ATR): ν 2052 (m), 1902 (s), 1555 (w), 1500 (w), 1448 (w), 1376 (w), 1188 (w), 1111 (w), 1068 (w), 1013 (w) cm^{-1} . Anal. Calcd for $\text{C}_{18}\text{H}_{12}\text{CrO}_7$: C, 55.11; H, 3.08. Found: C, 55.00; H, 3.17.

Pentacarbonyl[ethoxy[5-(2-phenylethynyl)-2-furyl]methylene]chromium(0) (7k). The general procedure B starting from BuLi (0.45 mL, 1.10 mmol), phenylacetylene (0.11 mL, 1.0 mmol), ZnBr_2 (0.225 g, 1.0 mmol), **6b** (0.435 g, 1.1 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, hexane then hexane/DCM 9/1) afforded 0.300 g of the title compound (72%) as a red solid. Mp: >96 °C dec. ^1H NMR (CDCl_3): δ 1.66 (t, J = 7.2 Hz, 3H, CH_3), 5.16 (q, J = 7.2 Hz, 2H, CH_2), 6.76 (d, J = 3.9 Hz, 1H, CH), 7.02 (d, J = 3.9 Hz, 1H, CH), 7.39–7.41 (m, 3H, CH), 7.57–7.61 (m, 2H, CH). ^{13}C NMR (CDCl_3): δ 15.3, 75.8, 79.1, 97.3, 113.9, 117.7, 121.3, 128.5, 129.6, 131.9, 143.5, 162.7, 216.8, 224.2, 308.6. IR (ATR): ν 2058 (m), 1985 (m), 1909 (s), 1550 (w), 1492 (w), 1460 (w), 1357 (w), 1262 (w), 1196 (w), 1151 (w), 1018 (w) cm^{-1} . Anal. Calcd for $\text{C}_{20}\text{H}_{12}\text{CrO}_7$: C, 57.70; H, 2.91. Found: C, 57.51; H, 2.99.

Pentacarbonyl[ethoxy(5-phenyl-2-thienyl)methylene]tungsten(0) (7l). The general procedure B starting from PhMgCl (0.60 mL, 1.0 mmol), ZnBr_2 (0.225 g, 1.0 mmol), **6c** (0.597 g, 1.1 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, hexane then hexane/DCM 4/1) afforded 0.466 g of the title compound (86%) as a red solid. Mp: >119 °C dec. ^1H NMR (CDCl_3): δ 1.67 (t, J = 6.6 Hz, 3H, CH_3), 4.99 (q, J = 6.6 Hz, 2H, CH_2), 7.42–7.46 (m, 4H, CH), 7.70–7.72 (m, 2H, CH), 8.15–8.17 (m, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.0, 78.2, 125.0, 126.3, 129.2, 129.7, 133.1, 143.4, 155.0, 156.3, 197.7 ($J_{183\text{W}-13\text{C}}$ = 127 Hz), 202.5, 287.7. IR (ATR): ν 2058 (s), 1963 (w), 1882 (s), 1520 (w), 1429 (m), 1346 (w), 1249 (m), 1180 (m), 1109 (w), 1063 (w), 1011 (w) cm^{-1} . Anal. Calcd for $\text{C}_{18}\text{H}_{12}\text{O}_6\text{SW}$: C, 40.02; H, 2.24; S, 5.94. Found: C, 40.19; H, 2.27; S, 5.96.

Pentacarbonyl[ethoxy[5-(2-phenylethynyl)-2-thienyl]methylene]tungsten(0) (7m). The general procedure B starting from BuLi (0.45 mL, 1.10 mmol), phenylacetylene (0.11 mL, 1.0 mmol), ZnBr_2 (0.225 g, 1.0 mmol), **6c** (0.597 g, 1.1 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, hexane then hexane/DCM 4/1) afforded 0.285 g of the title compound (51%) as a red solid. Mp: 79–81 °C. ^1H NMR (CDCl_3): δ 1.66 (t, J = 7.2 Hz, 3H, CH_3), 4.98 (q, J = 7.2 Hz, 2H, CH_2), 7.30–7.38 (m, 4H, CH), 7.53–7.56 (m, 2H, CH), 8.04–8.05 (m, 1H, CH). ^{13}C NMR (CDCl_3): δ 14.9, 78.4, 82.8, 99.2, 122.0, 128.5, 129.4, 131.6, 133.0, 133.4, 141.1, 157.8, 197.4 ($J_{183\text{W}-13\text{C}}$ = 127 Hz), 202.4, 288.5. IR (ATR): ν 2064 (m), 1975 (w), 1889 (s), 1481 (w), 1414 (m), 1344 (w), 1256 (w), 1191 (m), 1159 (m), 1108 (w), 1015 (w) cm^{-1} . Anal. Calcd for $\text{C}_{20}\text{H}_{12}\text{O}_6\text{SW}$: C, 42.58; H, 2.14; S, 5.68. Found: C, 42.79; H, 2.24; S, 5.66.

Pentacarbonyl[ethoxy[5-phenyl-2-thienyl]methylene]molybdenum(0) (12a). The general procedure B starting from BuLi (0.45 mL, 1.10 mmol), bromobenzene (0.157 g, 1.0 mmol), ZnBr_2 (0.225 g, 1.0 mmol), **10** (0.500 g, 1.1 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, hexane) afforded 0.350 g of the title compound (77%) as a red solid. Mp: >81 °C dec. ^1H NMR (CDCl_3): δ 1.66 (t, J = 6.9 Hz, 3H, CH_3), 5.06 (q, J = 6.9 Hz, 2H, CH_2), 7.41–7.46 (m, 4H, CH), 7.69–7.72 (m, 2H, CH), 8.19 (d, J = 4.5 Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 15.1, 77.5, 124.9, 126.3, 129.2, 129.7, 133.0, 143.3, 154.0, 155.0, 206.2, 212.9, 303.7. Anal. Calcd for $\text{C}_{18}\text{H}_{12}\text{MoO}_6\text{S}$: C, 47.80; H, 2.67; S, 7.09. Found: C, 48.13; H, 2.85; S, 7.57.

Pentacarbonyl[ethoxy[5-(2-phenylethynyl)-2-thienyl]methylene]molybdenum(0) (12b). The general procedure B starting from BuLi (0.45 mL, 1.10 mmol), phenylacetylene (0.11 mL, 1.0 mmol), ZnBr_2 (0.225 g, 1.0 mmol), **10** (0.597 g, 1.1 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, hexane) afforded 0.370 g of the title compound (78%) as a red wax. ^1H NMR (CDCl_3): δ 1.67 (t, J = 6.9 Hz, 3H, CH_3), 5.07 (q, J = 6.9 Hz, 2H, CH_2), 7.33 (d, J = 4.2 Hz, 1H, CH), 7.39–7.41 (m, 3H, CH), 7.55–7.59 (m, 2H, CH), 8.11 (d, J = 4.2 Hz, 1H, CH). Anal. Calcd for $\text{C}_{20}\text{H}_{12}\text{MoO}_6\text{S}$: C, 50.43; H, 2.54; S, 6.73. Found: C, 50.66; H, 2.70; S, 7.02.

Pentacarbonyl[(*N,N*-dimethylamino)(2-thienyl)methylene]molybdenum(0) (9). Butyllithium (8.2 mL, 20.0 mmol) was added to a solution of thiophene (1.60 g, 20.0 mmol) in dry THF (40 mL)

cooled to -78°C . The resultant mixture was stirred for 30 min at 0°C , then $\text{Mo}(\text{CO})_6$ (5.28 g, 20.0 mmol) was added and the mixture was stirred for another 2 h at ambient temperature. The solvents were then removed under reduced pressure, and the isolated oily residue was dissolved in water (40 mL) and cooled to 0°C with ice–water bath. Then Et_3OBF_4 (5.70 g, 30.0 mmol) was added, the reaction mixture was stirred for 10 min at 0°C , and precipitated red solid was isolated by extraction with ethyl acetate (3×100 mL). Combined organic layers were dried (MgSO_4), dry solution of the alkoxycarbene was mixed with dimethylamine (20 mL, 40 mmol, 2 M solution in ether), and the resultant mixture was stirred 2 h at ambient temperature. Then the solvents were removed under reduced pressure and column chromatography (silica gel, DCM/hexane 1/4 then 1/2) gave 4.56 g of the title compound (61%) as a yellow solid. Mp: $>81^{\circ}\text{C}$ dec. ^1H NMR (CDCl_3): δ 3.23 (s, 3H, CH_3), 3.95 (s, 3H, CH_3), 6.54 (dd, $J = 1.2, 3.6$ Hz, 1H, CH), 6.99 (dd, $J = 3.6, 3.9$ Hz, 1H, CH), 7.31 (dd, $J = 1.2, 3.9$ Hz, 1H, CH). ^{13}C NMR (CDCl_3): δ 45.1, 53.0, 118.6, 124.3, 126.9, 153.0, 205.9, 214.4, 262.5. Anal. Calcd for $\text{C}_{12}\text{H}_9\text{CrNO}_6\text{S}$: C, 38.41; H, 2.42; N, 3.73; S, 8.55. Found: C, 38.86; H, 2.53; N, 3.81; S, 8.80.

Pentacarbonyl[(*N,N*-dimethylamino)[5-(4-acetylphenyl)-2-thienyl]methylene]molybdenum(0) (11a). The general procedure A starting from **9** (0.375 g, 1.0 mmol), BuLi (0.53 mL, 1.3 mmol), ZnBr_2 (0.225 g, 1.0 mmol), 4-iodoacetophenone (0.492 g, 2.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.280 g of the title compound (57%) as a yellow solid. Mp: $>89^{\circ}\text{C}$ dec. ^1H NMR (CDCl_3): δ 2.61 (s, 3H, CH_3), 3.32 (s, 3H, CH_3), 3.98 (s, 3H, CH_3), 6.55 (d, $J = 3.9$ Hz, 1H, CH), 7.32 (d, $J = 3.9$ Hz, 1H, CH), 7.65 (d, $J = 8.7$ Hz, 2H, CH), 7.97 (d, $J = 8.7$ Hz, 2H, CH). ^{13}C NMR (CDCl_3): δ 26.5, 45.3, 52.9, 119.9, 124.5, 125.2, 129.1, 135.8, 137.9, 141.8, 153.3, 197.2, 205.8, 214.0, 261.5. Anal. Calcd for $\text{C}_{20}\text{H}_{15}\text{MoNO}_6\text{S}$: C, 48.69; H, 3.06; N, 2.84; S, 6.50. Found: C, 48.97; H, 3.55; N, 2.56; S, 6.50.

Pentacarbonyl[(*N,N*-dimethylamino)[5-(4-formylphenyl)-2-thienyl]methylene]molybdenum(0) (11b). The general procedure A starting from **9** (0.375 g, 1.0 mmol), BuLi (0.53 mL, 1.3 mmol), ZnBr_2 (0.225 g, 1.0 mmol), 4-benzaldehyde (0.464 g, 2.0 mmol) and $\text{Pd}(\text{PPh}_3)_4$ (0.058 g, 0.05 mmol) followed by column chromatography (silica gel, DCM/hexane 1/2 then 2/1) afforded 0.256 g of the title compound (53%) as a yellow solid. Mp: $>90^{\circ}\text{C}$ dec. ^1H NMR (CDCl_3): δ 3.31 (s, 3H, CH_3), 3.96 (s, 3H, CH_3), 6.55 (d, $J = 3.3$ Hz, 1H, CH), 7.35 (d, $J = 3.3$ Hz, 1H, CH), 7.69 (d, $J = 8.7$ Hz, 2H, CH), 7.86 (d, $J = 8.7$ Hz, 2H, CH), 9.97 (s, 1H, CH). ^{13}C NMR (CDCl_3): δ 45.3, 52.9, 119.9, 124.9, 125.6, 130.4, 135.1, 139.2, 141.5, 153.7, 191.3, 205.7, 214.0, 261.1. Anal. Calcd for $\text{C}_{19}\text{H}_{13}\text{MoNO}_6\text{S}$: C, 47.41; H, 2.73; N, 2.92; S, 6.69. Found: C, 48.03; H, 3.10; N, 2.72; S, 6.92.

■ ASSOCIATED CONTENT

Supporting Information

^1H and ^{13}C NMR spectra copies. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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